STICKIT FREESTYLE SCORING

Complete User Guide

64 topics covering every StickIt feature

For: Officials, judges, admins, and stream crews

StickIt v2.3.00

Contents

Getting Started

Welcome

StickIt is a full-stack scoring application for freestyle ski competitions sanctioned by US Ski & Snowboard (USSS) and FIS. It runs single-mogul, dual mogul, and aerials events for everything from a small Devo or RQS-EQS qualifier up through national and FIS-sanctioned championships.

This guide is the in-app reference for everyone who touches the app:

- **Officials** — chief of competition, scoring operators, registrars, TDs
- **Judges** — T&L, Air, Aerials, Dual Mogul, Timekeepers, Head Judges working from tablets
- **Admins** — system administrators managing users, backups, and the master databases
- **Public** — spectators reading the live scoreboard, broadcasters wiring up the OBS overlay

How to use this guide

The left sidebar groups topics by audience. Click a group header to expand or collapse it. Click a topic to read it. Use **Previous** and **Next** at the bottom of every topic to walk through a section in order.

The **search box** in the top bar filters the sidebar in real time. It matches topic titles, group labels, and the body of each topic — so a search for `bib`, `pace time`, or `aerials reduction` will narrow the list to just relevant pages.

You can deep-link to any topic by URL — for example, /help/quick-start or /help/scoring-voice — which makes it easy to send a teammate the exact instructions they need.

What's not in this guide

- **Step-by-step training videos** — text only at the moment.
- **Internal architecture or developer notes** — those live in the project's `CLAUDE.md` and `CHANGELOG.md` files.
- **Live troubleshooting** — for help during a meet, contact the Rocky Mountain Freestyle scoring team or open an issue against the StickIt project.

Where to start

If you're brand new to StickIt, read [App overview & user roles]() next, then [Quick start: Run your first meet](). Both are short.

If you came here looking for one specific feature, use the search box — every page in this guide is indexed.

App overview & user roles

StickIt is split into four zones, each with its own URL space and visual style. Knowing which zone you're in tells you which sidebar and which permissions apply.

The four zones

Zone	URL	Sidebar	Who uses it
Public	`/`, `/livescores`, `/scoreboard/...`, `/overlay/...`	None — full-screen	Spectators, broadcasters, anyone with the link
Officials	`/dashboard/...`	Officials sidebar (Meets, Athletes, USSS Database)	Chief of competition, scoring operators
Tablets	`/judge/...`, `/aerials-judge/...`, `/headjudge/...`, `/timekeeper/...`	None — full-screen tablet UI	Judges and Timekeepers, one tablet per role
Admin	`/admin/...`	Admin sidebar (Dashboard, Users, Events, etc.)	System administrators

A meet director typically lives in the Officials zone all day, only opening Admin to handle a backup or a user account.

What each role does

Officials create meets, add events, register athletes, build the run order, generate tablet URLs for the judges, run the live scoring, finalize results, and produce PDFs / CSV exports for the TD report and USSS submission. The bulk of this guide is for them.

Judges open one URL on a tablet at the start of the day and submit scores run after run. The URLs carry unguessable short codes — no login needed, and judge tablets keep working even when password protection is on. They never see the meet setup or the master databases. Each judge tablet is locked to one event and one role.

Admins look after the system: managing login accounts and password protection, locking events or hiding test events from Live Scores, syncing the USSS People database, viewing the audit log, and managing automatic backups.

The public doesn't sign in. They open `/livescores` to see what's happening across the system, click through to a `/scoreboard/<short>` for a specific event, or watch the `/overlay/<short>` broadcast view fed into OBS or a YoloBox.

Roles within scoring

Inside a single event, scoring breaks down further:

- **T&L Judges** (3 for Comp Series, 2 for Devo / RQS-EQS) — score Carving, Absorption / Extension, Upper Body, and Deduction (or a single raw score in non-component mode).
- **Air Judges** (2 for Comp Series, 1 for Devo / RQS-EQS) — score each jump per athlete, also enter the jump codes.
- **Aerials Judges** (panel of 2–7) — score Air, Form, and Landing per jump per judge.

- **Timekeeper** — enters run time after each athlete finishes.
- **Head Judge** — reviews and approves each athlete's full score, can send back individual judge submissions for re-entry, and finalizes the event at the end.

The next topic, Quick start: Run your first meet, walks the smallest possible end-to-end flow.

Quick start: Run your first meet

This is the smallest end-to-end flow — create a meet, add an event, register a few athletes, score them, and publish results. About 20 minutes from start to finish.

1. Create a meet

1. From the home page click **Officials** to enter the dashboard.
2. Click **+ New Meet**.
3. Enter a name, location, and date. Save.

You're now on the meet detail page.

2. Add an event

1. Click **+ Add Event**.
2. Choose **Mogul** discipline, **Comp Series** category, **Male** gender. The event auto-names itself "Comp Series Male Mogul".
3. Save. The event card appears.

3. Set course specifications

Scroll down on the meet page to **Course Specifications**. Enter a course length (e.g. `220` meters) and confirm the pace standard. **USSS** is the default (9.70 m/s men, 8.20 m/s women). Pace times calculate automatically. See Course specifications for details.

4. Click into the event

Click the event card to open the event detail page. You'll see tabs at the top: **Setup**, **Registration**, **Scoring**, **Results**, **Phases**, **Links**, etc.

5. Register athletes

Go to the **Registration** tab. The fastest path:

1. Click **+ Add Athlete** and use the **From USSS Database** picker if you've synced the USSS People file (see USSS sync). Otherwise click **Manual Entry**.
2. Add at least 2 athletes. Each needs a first name, last name, USSS#, birth year, and bib number.
3. The Run Order section appears once registrations exist. Click **Random Order** to shuffle, or **By Age Groups** to group U-class first.
4. Click **Save Order**.

6. Add judges

Go to the **Setup** tab.

1. Click **+ Add Judge** and add 3 T&L judges (TL1/TL2/TL3) and 2 Air judges (Air1/Air2). Give each a name and a PIN.
2. Click **Links**. Each judge has a unique tablet URL. Open the T&L1 URL in a separate browser tab to simulate a judge tablet.

7. Score one run

1. Back on the **Scoring** tab, click **Start Run** for the first athlete.
2. On each judge tablet, enter scores for the run. Submit.
3. Once all judges have submitted (and the timekeeper has entered a time), the Head Judge tablet shows the full score for review.
4. On the HJ tablet, click **Approve & Submit**.
5. The run is finalized — the score appears on the **Results** tab and on the public scoreboard.

8. Publish results

Click **Reports** (or the PDF icons in the **Results** tab) to generate the **Event Results Summary** PDF. That's the document you hand to the TD.

Where to go next

- Reading the results table explains what every column means.
- Live scoring flow covers the full live-scoring loop in more detail.
- Importing from SkiReg CSV is the production workflow for registering 30+ athletes at once.

Officials — Meets

Creating a meet

A **meet** is the top-level container for everything that happens on a competition weekend. One meet can hold many events — for example, a single weekend might be one meet with six events: men's mogul, women's mogul, men's dual, women's dual, men's aerials, women's aerials.

Steps

1. Go to the **Officials** dashboard (the home of the Meets list).
2. Click **+ New Meet** in the top-right of the page.
3. Fill in: - **Meet name** — the public-facing name. This shows on the scoreboard, on the live-scores page, and on every PDF. Use the official name from the registration site (e.g., `2026 Winter Park Spring Series #3`). - **Location** — usually the resort or venue (`Winter Park, CO`). - **Start date** — the first day of competition. If the meet runs more than one day, this is day one.
4. Click **Create**. You're taken to the meet detail page.

What happens behind the scenes

A new row is created in the `meets` table with a generated UUID and a unique short code. The short code is what appears in public URLs like `/scoreboard/abc12` so spectators don't have to type a UUID.

The meet starts empty — no events, no athletes, no course specs. Add those next:

- Adding an event
- Course specifications
- Registering athletes for an event

Field-naming conventions

USSS sanction documents tend to use names like `2026 USSA EC #2 Mogul Championships`. StickIt does no naming validation — type whatever your sanction shows. The meet name is preserved character-for-character in PDFs and exports, which simplifies cross-checking against the sanction.

Editing later

You can change any meet field after creation — see Editing meet settings. The meet name is safe to edit at any time; nothing in the database keys off it.

Deleting

If you create a meet by mistake, see Deleting a meet. Deletion cascades to every event, registration, run, score, judge, bracket match, phase, and run-order entry under that meet — so use it carefully.

Editing meet settings

Meet name, location, and start date can be changed at any time without affecting any scoring data underneath.

Steps

1. Open the meet from the dashboard.
2. Below the meet's location and date, click **Edit Meet Settings**.
3. The same modal you used to create the meet opens, pre-filled with current values.
4. Change what you need and click **Save**.

What you can edit

- **Meet name** — safe to edit anytime. The change propagates to the scoreboard, the live-scores listing, every PDF, and every export.
- **Location** — same as above.
- **Start date** — used by `By Age Groups` run-order builds to compute each athlete's USSS age class. Changing the date *after* you've built a run order will not retro-recompute the age classes; rebuild the run order if the date change crosses a USSS season boundary (July 1).

What you can't edit here

- **Discipline** of an event — that's an event property, not a meet property. See Editing an event.
- **Course length / pace standard** — those live on the meet's Course specifications panel.
- **The meet's short code** — auto-generated, stable for the life of the meet. Sharing the same `/scoreboard/<short>` link before and after an edit always works.

The Advanced panel

Next to **Edit Meet Settings** is an **Advanced** button, which opens the meet-level rule and permission settings. These apply to every event in the meet:

Setting	Default	Effect
Landing Past the Lower Chop (NJ) rule	Off	When off, the FIS FS-18 chop/NJ controls are hidden on the dual judge and Head Judge tablets, in the paper-score modal, and on the Scoring tab — and the server refuses NJ calls. Historical matches that already carry an NJ finding still display it.
Air score tie allowed	Off	When off, the dual Air Judge (J3) cannot submit a tied air score — the Air Tied button and checkbox are hidden and the server refuses the submission. Time Tied (J4) is unaffected.
Who can start a run	All on	Three independent checkboxes control whether the **Start Run** button (and its DNS companion) appears on the Timekeeper tablet, the Head Judge tablet, and the Scoring tab. If all three are turned off, the Scoring tab keeps its button anyway so the meet can never be locked out.

Setting	Default	Effect
Allow venue server adoption	On	When off, this meet is cloud-only (remote judging) and can never be released to or adopted by a venue server. Locks once the meet is adopted. This control used to live in Edit Meet Settings as "Remote judging meet".

All four settings survive meet export/import and cloning, and (except the adoption flag) ride along when a venue server adopts the meet.

The More menu

The meet header groups its less-frequent actions — **TD Report**, **Export Meet**, **Clone Meet** — under a **More** button to keep the header compact.

Status changes

The meet and event **Status** dropdowns ask for confirmation when changing to or from **Complete**, since manually completing an event bypasses the Head Judge finalize flow.

Visibility

Edits to a meet's name or date are immediate. Officials viewing the dashboard at the same moment will see the change on their next page-refresh. Public scoreboards refresh automatically — anyone watching `/scoreboard/<short>` sees the new title within a few seconds via the live polling cycle.

Course specifications

The **Course Specifications** panel on the meet page sets the course length and pace standard. These two values drive every speed-score calculation for every mogul run at the meet.

What gets stored

Each meet has one course spec record:

- **Course length** — in meters. Measured from the start gate to the finish line.
- **Pace standard** — `USSS` (default) or `FIS`. Determines the pace speeds used to calculate pace time:
- **USSS** — Men 9.70 m/s, Women 8.20 m/s
- **FIS** — Men 10.30 m/s, Women 9.00 m/s
- **Pace time (men)** and **pace time (women)** — computed automatically as `course length / pace speed`, truncated to two decimals per FIS rule.

Setting it up

1. On the meet detail page, scroll to **Course Specifications**.
2. Enter the course length.
3. Click **USSS** or **FIS** to pick the standard. The pace times update live as you toggle.
4. Click **Save**.

The values then propagate down to every existing event under the meet via `propagatePaceToEvents`. New events created later inherit the meet's spec automatically (`inheritCourseSpec`).

Per-event overrides

In rare cases an event needs a different pace time — usually because the course was shortened mid-day for weather. Set a per-event `pace_time_override` from the event's Setup tab; that value will be used in the speed-score formula for that event only, ignoring the meet-level pace.

How speed score uses these values

The mogul speed score formula is:

```
speedScore = max(0, 48 - 32 × (run_time / pace_time))
```

then clamped to the 20-point cap and floored to two decimals. So pace time directly determines the curve. A longer course › larger pace time › easier to hit the 20-point ceiling. A shorter course › smaller pace time › tougher curve.

When to enter zero-length

If you don't yet know the course length on Day 1 morning, leave the field blank and enter `0`. Speed scores will all calculate to `0.00` until you fill in a real length and click **Save** — at which point every completed run is recomputed via `recalcSpeedScores`. (This rewrites `runs.total_score` for every finalized run, then broadcasts updated standings.)

Pace standard choice

Use **USSS** for any USSS-sanctioned event. Switch to **FIS** only for FIS-sanctioned competitions (NorAm, World Cup, etc.). The choice is preserved per meet and printed on the calculation report so the TD can verify.

Training days

A **training day** is a participant roster for a pre-comp on-hill training session. Most meets have one or more training days (e.g. Thursday before a Saturday/Sunday comp). StickIt lets you build one or more named training days per meet, print attendance rosters as PDFs, and refine each day's list without touching event registrations.

Opening Training Days

From a meet's detail page (Dashboard › Meets › *meet name*), click the **Training Days** button in the action row at the top. You land on the standalone Training Days page for that meet.

Creating a training day

Click **+ New** in the left panel. Enter:

- **Name** — free text, e.g. "Thursday Training" or "Training Day 1".
- **Date** *(optional)* — used for sorting and printed on the PDF header.

Click **Create**. The new day appears in the left list and is auto-selected.

Default participant list

When you select a training day, the right panel shows every athlete registered across every event in this meet — deduplicated by athlete, with the first non-empty bib number shown. The list starts with every athlete **included** ([yes] in the leftmost column). Athletes whose event registration is set to **Scratched** are excluded by default and don't appear at all.

Column	Source
[yes]	Inclusion toggle (saved per training day)
Bib	First non-empty bib across the athlete's event registrations
Athlete	`Last, First` from the master athletes record
USSA #	USSS membership number
Club	Athlete's club

Opting athletes out

Uncheck the box on any row to exclude that athlete from this specific training day. The change saves immediately. Exclusions are scoped to one training day only — checking the same athlete back in works the same way. Exclusions are stored per `(training_day_id, athlete_id)` pair.

Use the header master checkbox to **include all** or **exclude all** at once. The **Reset (Include All)** link clears every exclusion on the current day.

Live syncing with registrations

The list is computed live from event registrations every time you open the page. New athletes registered to any event under this meet AFTER the training day was created **automatically appear** in the list (checked). Athletes removed from event registrations automatically disappear. Scratching an athlete in an event also removes them.

This means you don't need to manually maintain training day lists across registration changes — just open the day and the list is current.

Printing the PDF

Click **Print PDF** in the right-panel header. The PDF lists only included athletes, sorted by bib (blanks last), then by last name. Columns: Bib · Last · First · USSA # · Club. The header shows the meet name plus `Training Day Participants — <day name> — <formatted date>`. Filename: `Training_Day_<MeetSlug>_<DaySlug>.pdf`.

Pin the printout at the training day check-in table — coaches mark who showed up.

Multiple training days

Create as many days as you need on the same meet. Each has its own independent inclusion/exclusion state. Switching between days in the left list does not affect the others. Days are sorted by date ascending (blanks last), then by creation time.

Editing or deleting a day

Hover any day in the left list — the (edit) button opens an edit modal for name/date; the x button deletes the day after confirmation. Deleting a training day removes its exclusion records but does not affect any event registrations.

When the meet is deleted

Training days and their exclusion lists are removed along with the meet. There's nothing to clean up manually.

Included in meet export/import

Exporting a meet (Exporting a meet) includes its training days and each day's participant exclusion list, and they survive the round-trip: a fresh import recreates them exactly, and merging an updated export into an existing meet matches training days by name, keeps the newer date, and never loses exclusions.

Importing a meet from file

Meet files (`.zip` exports from another StickIt installation) can be imported back into any StickIt server. This is the standard way to:

- Move a meet from a practice server to the production server
- Restore a meet from a hand-saved backup
- Bring a partner organization's meet into your system

Steps (single meet)

1. From the dashboard, click **Import Meet**.
2. Select a `.zip` or `.json` file from your computer.
3. The server reads it and inspects for naming conflicts.
4. **No conflict** — the meet imports immediately. You're returned to the dashboard with the new meet at the top.
5. **Conflict** — a meet with the same name already exists. A modal appears with side-by-side comparison cards (server version vs. import file) and four options: - **Cancel** — abort. Nothing changes. - **Import as Duplicate** — creates a new meet with `(Duplicate)` appended to the name. If that name also exists, uses `(Duplicate 2)`, etc. - **Merge** — adds new data and updates older data on the existing meet. See merge rules below. - **Overwrite** — requires a second confirmation. Deletes all existing data for the meet and replaces it with the import file's content.

Merge rules (when you choose Merge)

- Events match by `discipline + gender` (with name as a fallback).
- Athletes deduplicate by USSA#, FIS ID, or name+birth_year.
- Registrations match by event + athlete.
- Judges match by event + role.
- Runs match by event + athlete + run number + round.
- Judge scores match by run + judge + score type.
- For every match, the **newer** record (`updated_at` / `submitted_at`) wins. Older import data is ignored.

Merge is non-destructive — you can run it any number of times.

Multi-meet import (ZIP-of-zips)

If the file you uploaded was created via **Export All** (see Exporting a meet), it contains a `manifest.json` listing all bundled meets. The import flow then becomes:

1. A modal lists every meet in the bundle with checkboxes.
2. Conflict-flagged meets show an amber **Already exists** badge.
3. Pick which meets to import. Click **Import N meets**.
4. The flow runs them sequentially — for each conflict-flagged meet, the standard conflict modal appears so you can pick Merge / Duplicate / Overwrite per meet.

5. A summary at the end lists each meet's outcome.

Raw JSON files

The importer also accepts a bare `meet_export.json` file (no zip wrapping). This is mostly useful for hand-editing edge cases — the standard workflow uses zips.

What gets imported

Everything in the meet's scope: meet metadata, course specs, events with full configuration (event type, panel size, reduction method, phases, dual mogul random seed, lock state), athletes, registrations, judges, runs, judge scores, dual bracket matches, dual judge points, phase run order, run-round status. Audit log entries are *not* imported (those are server-local).

Exporting a meet

Every meet can be exported as a self-contained `.zip` file. The export includes the meet, every event, every registration, every run, every judge score, every dual bracket match, every phase, every training day (with its participant exclusion list), and the entire audit trail for that meet.

Single meet

1. Open the meet from the dashboard.
2. Click **Export Meet** in the meet header.
3. The browser downloads `meet_<short-code>.zip` (typically a few hundred KB to a few MB).

Export All

To bundle every visible meet on the server in one go:

1. From the dashboard, click **Export All**.
2. The server creates a **zip-of-zips**: one parent `.zip` containing a `manifest.json` plus one child `.zip` per meet.
3. The download is named `StickIt_AllMeets_YYYYMMDD.zip`.

`Export All` respects the dashboard's filter — fully-locked meets that are hidden from your dashboard are excluded automatically.

Bundle contents

```
StickIt_AllMeets_20260513.zip
|--- manifest.json
|--- meet_abc12.zip
|--- meet_def34.zip
`--- meet_ghi56.zip
```

The `manifest.json` lists each meet's `meet_id`, `name`, `location`, `date`, `meet_ranking`, and child filename. Each child zip contains a single `meet_export.json` with the full meet payload — byte-identical to a single-meet export.

Safari note

If you're on macOS Safari and the download arrives extracted (a folder instead of a `.zip`), make sure you're on v1.16.25 or newer. The export endpoints now send `Content-Type: application/octet-stream` so Safari treats the file as opaque binary and skips its auto-extract heuristic. The filename keeps the `.zip` extension so double-clicking still extracts normally via Archive Utility.

What's not in the export

- No images, audio, or PDFs — only the structured JSON.

- No master USSS People database (that's separate; see Admin › USSS People).
- No system-level audit log entries from before the meet was created.
- No user accounts.

Re-importing

Use the Importing a meet from file flow. Single-meet exports import as one meet (with the conflict dialog if one already exists). Multi-meet bundles open the multi-meet picker.

When to export

- **Before risky operations** — for example, before re-seeding a dual bracket on Day 2.
- **End of meet** — keep a permanent local copy of the final state for the season archive.
- **Sharing** — if a partner organization needs the data, an export is the cleanest hand-off.
- **Migrating** — when moving a meet to a different StickIt server (e.g., promoting from a practice instance to the production deploy).

Deleting a meet

Deleting a meet is permanent and **cascades**. Every event, registration, run, judge score, judge, dual bracket match, dual judge points, phase, phase run-order entry, run-round status entry, and audit log entry under that meet is removed.

Steps

1. Open the meet from the dashboard.
2. Click **Delete Meet** in the meet header.
3. A confirmation modal appears showing the meet name and **how many events it contains** — a quick sanity check that you're deleting the right meet.
4. The delete is two-step: arm the action first, then click the final destructive **Delete** button. You're returned to the dashboard.

What you should do first

Before deleting:

- **Export the meet** (Exporting a meet) so you have a permanent backup. The auto-backup system will also have a snapshot, but a fresh dedicated export is the safest path.
- **Check the audit log** (Audit log) if there's any chance the meet contains scores you need for a TD report.
- **Verify scores have been transmitted** to USSS if applicable — see USSS transmit XML.

Cannot-undo warning

There is no soft-delete on meets. Once you click confirm, the rows are removed from the database. Recovery requires either:

- Importing your local export back in (see Importing a meet from file)
- Restoring from an automated backup (see Backups)

When deletion is appropriate

- A meet was created by mistake (wrong date, wrong sanction, wrong year).
- A test meet with synthetic data is no longer needed.
- A meet has been finalized, all PDFs have been generated, USSS XML has been transmitted, and you're cleaning up old data.

When to use Lock instead

If your goal is "stop people from changing this meet" but you want to keep the data, use **Lock** at the event level (see Locking & unlocking events) or in the Admin events page. Locked events become read-only but the data stays intact and stays visible on public scoreboards.

Deletion does not affect

- The master Athletes database — athlete rows live independently of any meet.
- The USSS People database — system-wide.
- Backup files — those are separate.
- Other meets — only the meet you confirmed is removed.

Officials — Events

Adding an event
Editing an event
Categories & divisions explained
Multi-phase events
Dual mogul setup
Aerials event setup
Locking & unlocking events

Adding an event

An **event** is one competition within a meet — a single discipline + gender combination, often run as one or more phases on a given day.

Steps

1. Open the meet detail page.
2. Click **+ Add Event**.
3. The event form modal opens. Fill in: - **Discipline** — `Mogul`, `Dual Mogul`, or `Aerials`. - **Category** — varies by discipline: - **Mogul / Dual Mogul:** Comp Series (default), Devo, RQS-EQS, FIS. - **Aerials:** USA Regional (default), USA National, FIS Other, FIS NAC/NorAm, FIS OWG/WSC/WC. See [Categories & divisions explained] for what each one means and which judging configuration it implies. - **Gender** — `Male`, `Female`, or `Mixed`. - **Event name** — auto-fills from the discipline + category + gender (e.g. `Comp Series Male Mogul`). You can override. - **Date** — defaults to the meet's start date. Set explicitly for multi-day meets.
4. Click **Create Event**. The event appears as a card on the meet detail page.

What's set automatically

- For aerials, picking a **Category** sets the panel-size range, the HJ-may-score flag, and the reduction method options. See Aerials event setup.
- For mogul / dual mogul, picking a **Category** sets the default judging configuration:
- **Comp Series:** 3 T&L + 2 Air judges, 2 jumps, time on, component scoring on.
- **RQS-EQS:** 2 T&L + 1 Air, 2 jumps, time on, component scoring **off** (single raw T&L score).
- **Devo:** 2 T&L + 1 Air, 1 jump, **no time**.
- The event inherits the meet's Course specifications automatically.
- The event gets a unique short code for public URLs (e.g. `/scoreboard/abc12`).

Auto-naming

The auto-naming follows simple rules:

Discipline	Category	Format
Mogul / Dual Mogul	Any	`<Category> <Gender> <Discipline>` — e.g. `Comp Series Female Mogul`
Aerials	Any	`<Event Type Label> <Gender> Aerials` — e.g. `USA Regional Male Aerials`

You can edit the name to whatever you like (`Day 1 — Mens Comp`, `Saturday Mogul Final`, etc.). The auto-name only applies on first edit before you've changed it.

After creating

- Add judges on the **Setup** tab — see Adding & assigning judges.
- Register athletes on the **Registration** tab — see Registering athletes for an event.
- Configure phases if needed — see Multi-phase events.

- Generate tablet URLs from the **Links** tab — see Judge tablet URLs.

Editing an event

The **Edit** button appears on every event card on the meet detail page. Click it to re-open the same modal you used to create the event, pre-filled with current values.

When Edit is enabled

The Edit button is **disabled once the first run for the event has been recorded**. "First run" means any row exists in the `runs` table for the event — which happens when scoring starts (tablet flow) or when a manual entry is submitted.

When disabled, the button shows the tooltip:

> Cannot edit after scoring has started

This is to prevent operators accidentally changing structural fields (number of judges, jump count, component-scoring on/off) mid-meet, which would invalidate runs already in the database. If you absolutely must change something after scoring has started, you have a few options:

1. **Server-side guard fields** — even if you bypass the UI, the server's `PUT /events/:id` endpoint refuses changes to `component_scoring`, `score_entry_mode`, `event_type`, and `aerials_panel_size` once a complete run exists.
2. **Delete and re-create** — delete the event (cascades to all its runs) and re-create from scratch.
3. **Re-score the affected runs manually** after the structural change. Out of scope for the standard flow — use only with TD approval.

What you can safely edit pre-scoring

- Event name
- Date
- Category / division (within the same discipline)
- Number of T&L judges, Air judges, jumps
- Component-scoring on/off
- Has-speed on/off
- Aerials panel size, reduction method, HJ-may-score
- Pace time override

What you can never edit after creation

- **Discipline** — switching mogul ‹› dual mogul ‹› aerials would invalidate the entire event's data model. To change discipline, delete and re-create.

Saving

Click **Save Changes**. The event card updates immediately. No event-level data is reset; only the fields you changed are persisted.

Visibility

Other officials viewing the meet detail page see the change on next refresh. Public scoreboards show updated event names within a few seconds via the live polling cycle.

Categories & divisions explained

The Category dropdown in the Add Event modal does two things: it sets sane judging defaults and (for aerials) it picks the sanction tier that controls panel rules. Pick wrong and you'll spend the morning fighting the defaults.

Mogul / Dual Mogul categories

Category	Default config	Use for
Comp Series	3 T&L + 2 Air judges, 2 jumps, time on, component scoring on	All standard USSS competitions; most divisional / regional events
Devo	2 T&L + 1 Air judges, 1 jump, **no time**	Development events, U7–U13 grassroots; faster scoring, fewer fields
RQS-EQS	2 T&L + 1 Air judges, 2 jumps, time on, component scoring **off**	RQS / EQS series — single raw T&L score per judge instead of carving / absorption / upper-body breakdown
FIS	3 T&L + 2 Air judges, 2 jumps, time on, component scoring on	FIS-sanctioned (NorAm, World Cup)

The defaults are just starting points — you can override any field in the Add Event modal or via Editing an event before scoring begins.

Component vs. non-component T&L

When **component scoring is on**, each T&L judge enters four values per run:

- **Carving** (0–10)
- **Absorption / Extension** (0–5)
- **Upper Body** (0–5)
- **Deduction** (0–6, common values: 0.1 / 0.5 / 1.0 / 1.6 / 6.0)

Total per judge = `Carving + Absorption + Upper Body - Deduction`, clamped to 0.1–20.

When **component scoring is off** (RQS-EQS), each T&L judge enters two values:

- **Raw score** (0–20)
- **Deduction** (0–6)

Total per judge = `Raw - Deduction`, clamped to 0.1–20.

The Turns score (max 60) sums the three counting judge totals (5-judge format) or drops the high+low and sums the middle three (7-judge format), per FIS JH 6203.

Aerials event types (sanction tier)

Event Type	Panel Size	HJ may score?
FIS OWG/WSC/WC	5–7	No
FIS NAC/NorAm	5–7	No
FIS Other	5 (locked)	No

Event Type	Panel Size	HJ may score?
USA National	2–5	No
USA Regional (default)	2–5	Yes (HJ counts as a scoring judge)

These come from FIS Judging Handbook 6004 / USSS 4110. Picking the right event type unlocks the right panel-size range and reduction-method dropdown. See Aerials event setup.

Why this matters for results

Categories don't change the scoring math itself — but they change which fields the judges enter and how many judges score per run. Changing category mid-meet would invalidate any runs already submitted, which is why Editing an event refuses structural changes once scoring begins.

Multi-phase events

A **phase** is a distinct round of scoring within an event. The simplest event has one phase (one run, one set of scores). The more complex layouts:

- **Best of 2** — every athlete gets two runs. Final ranking uses the better of the two.
- **Qualifier / Finals** — Run 1 is a qualifier; the top N advance to Run 2 (the final). Optionally a third "Final 2" runoff.
- **Best of 2 with multi-finals** — combinations of the above.

Configuring phases

1. On the event detail page, click the **Phases** tab.
2. Click **Configure Phases**.
3. Pick a phase template: - **Single Run** (default — no phases configured) - **Best of 2** — Run 1 + Run 2, both score, take the better - **Qualifier / Finals** — Q1 + F1, with a configurable cut size (e.g. top 16 to F1) - **Qualifier / Two Finals** — Q1 + F1 + F2, with two cuts
4. For each cut, set the **Final size** (e.g. `16`).
5. Click **Save Phases**.

Phases generate a `phase_run_order` for each phase. Run 1 always uses the registration run order. Subsequent phases use the previous phase's results to determine who advances.

Cut-line tie expansion

Per FIS ICR 4207.3.4, when athletes tie at the unbreakable level (Total › Turns › Air-no-DD › Speed all equal) and the shared rank straddles a cut line, **all** of them advance. So a 16-athlete final can become a 17-athlete final if there's a two-way tie at rank 16. This is rules-correct behavior and happens silently — the cut just expands to include the tied athletes.

Undersized field warning

If you set a cut to 16 but only 8 athletes qualified (heavy DNS, last-minute withdrawals), the phase is created with whatever athletes are available — but a yellow notice appears on the Phases tab:

> Final 1 was configured for 16 athletes, but only 8 qualified.

Non-blocking; the meet continues normally. The notice is informational so the operator can confirm with the TD whether to re-open registration or proceed.

Q2 field limit (v1.26.00, WC Phased Finals)

The Spring 2026 FIS World Cup **Phased Finals** formats restrict Qualification 2 to a ranked band: only athletes ranked from `pass-through + 1` down to a fixed rank take a Q2 run; everyone below that rank is finished and ranks on their Q1 score in the qualification tier.

When adding a **Qualifier 2** phase, the optional **"Q2 field limit (last rank eligible for Q2)"** input sets that cap. Blank = no cap (every non-pass-through athlete runs Q2 — the Championship format and all USSS events). Ties at the limit rank expand the field per ICR 4207.3.4, the same as every other cut in StickIt.

Format presets for manual configuration:

Format	Pass-through	Q2 field limit	F1 size	F2 size
WC Phased Finals — Moguls	8	32	16	6
Championship — Moguls	10	none	20	8
WC Phased Finals — Aerials	6	18	12	6
Championship — Aerials	6	none	12	(two-jump F1 handled procedurally)

Run order across phases

- Run 1 / Q1 › registration run order (the order you saved in Building run order).
- Run 2 (Best of 2) › reverse order of Run 1 results (lowest scoring first, highest scoring last) per FIS ICR 4205.
- F1 / F2 › reverse order of the qualifier results, restricted to the cut.

Editing phases mid-event

Once a phase has been finalized you cannot reconfigure it from the UI. To re-open a phase:

1. The Head Judge taps **Send Back to Scoring** on their tablet (see Head Judge review & approval).
2. Edit individual scores via Editing a finalized score.
3. Re-finalize via the HJ tablet's Approve & Submit flow.

For non-Best-of-2 single-run events, no phase configuration is needed at all.

Dual mogul setup

Dual mogul is a head-to-head bracket format. Two athletes ski simultaneously on parallel courses; five judges award points on a 5-point split (e.g., a `4-1` decision means one athlete won 4 points to 1).

Bracket sizes

Pick from 4, 8, 16, 32, or 64 athletes. The bracket size must be >= the number of athletes you've registered.

- **4** — semis + final + 3rd/4th
- **8** — quarters + semis + final + 3rd/4th
- **16, 32, 64** — same pattern, one extra round each

Optional fields

- **Runoff to 8th** — if checked, the bracket runs a true 5–8 mini-bracket per USSS 4310.3.2: the four quarterfinal losers meet in two **consolation semifinals**, whose winners play the **5/6 final** and losers the **7/8 final** — in addition to the standard 3rd/4th. The finals run in order 7/8 › 5/6 › 3rd/4th › Championship at the end of the event. Useful when FFSP points need to extend down to 8th place.
- **Random seed (optional)** — used by the **Random Seed** seeding button to make seedings reproducible. Leave blank for true randomness; set a value if you want to be able to re-seed identically later.

Seeding the bracket

Once athletes are registered, go to the **Bracket** tab on the event detail page. Three seeding options:

1. **FIS Bracketing** — uses placement from another mogul event (typically the qualifier mogul event of the same gender from the same meet) to seed the bracket per the FIS dual mogul placement spec. Includes band-based randomization to avoid having seeds 1 and 2 always meet in the final and seeds 9 and 16 always meet in round one. See `server/dual/placement.js` for the algorithm.
2. **Random Seed** — shuffles all registered athletes randomly and lays them into the bracket positionally.
3. **Manual Bracketing** — opens a hand-placement modal with first-round match cards. Each card has Blue and Red dropdowns showing available athletes plus a BYE option. Place each athlete and BYE explicitly. Validation prevents both-BYE matches.

All three produce the same bracket-shell shape — once seeded, the live scoring flow is identical regardless of how the bracket was built.

Round-robin scoring

For each match:

1. The current match shows on the Scoring tab and on the dual mogul judge tablets.
2. Each of the 5 dual mogul judges (DualTurns1, DualTurns2, DualAir, DualTime, DualOverall) submits a 5-point split.
3. The HJ approves the match.

4. The winner advances to the next round; the loser may go into the consolation bracket.
5. The bracket re-renders with the next match queued.

End of bracket

Once every match (including consolation) is complete, the HJ tablet shows the final bracket tree with **Approve & Finalize Event** and **Send Back to Scoring** buttons. Approve to mark the event complete and unlock FFSP points (see FFSP points (dual mogul)).

Manual score entry

While a tablet-scored match is in progress, the operator can intercept with **Manual Score Entry** from the Scoring tab. This locks the judge tablets, opens a 5-judge entry modal, and lets the operator finalize the match manually (e.g. when a tablet is offline or scores are coming in over the radio). See Manual score entry.

Bottom air landing zone — the "chop" rule (NJ)

This rule is off by default. It is enabled per meet in the meet page's **Advanced** panel ("Landing Past the Lower Chop (NJ) rule") — the decision for the domestic season is not to use it, so unless a meet turns it on, every NJ control described below is hidden and the server refuses NJ calls. Historical matches that carry an NJ finding still display it everywhere.

Per the Spring 2026 FIS rule (FS-18), each competitor must land the **bottom air within the Landing Zone** (maximum 20 meters from the takeoff, control gates at 20.5 meters). A competitor whose boots or body land past the mark receives **No Jump (NJ) on the bottom air and zero speed points**. If **both** competitors land past, they are tied for speed and both receive NJ.

Who makes the call: the **Air Judge (J3)** — their dual tablet has an **NJ (Past Chop)** panel with independent Blue/Red toggles (each with a confirmation, on set and on clear). The **Head Judge** can also set or clear the call from the HJ tablet, and the flags are editable on the operator's Scoring tab and in the paper-score modal. The finding is locked once the HJ approves the match. The NJ call is independent of the air split — the Air Judge still submits a normal 5-point air comparison.

What the call does to the score (applied automatically at calculation time — the Time Judge's real entry is always recorded as evidence and never modified):

- **One competitor NJ** — the speed comparison pays **0 to the violator / 5 to the opponent**, regardless of what the Time Judge entered (including a Time Tied entry). The Overall Judge splits the normal 5 points.
- **Both NJ** — the speed comparison is a true tie: each side is credited **3 / 3** and the Overall Judge drops to a **4-point split**. (The 2.5 figure in the FIS text is the seven-judge panel value; StickIt's five-judge panel uses 3 / 3 per FIS JH 6304.3.2.)

The Time Judge's tablet shows a persistent amber banner while an NJ call is active. The Head Judge's approval view shows the finding, the Time Judge's recorded entry, and the effective (overridden) speed values side by side — approving the match certifies the finding. NJ badges appear on the athlete cards, the match rows, the public scoreboard, the broadcast overlay, and the bracket PDF (`[NJ]` name tag).

Tied speed and tied air

- **Time Tied** (J4 button/checkbox) — the speed comparison is tied. Since v1.29.00 the tied credit is **3 / 3** (previously displayed 0 / 0), so a tied-time match totals 25 points, not 19 — including historical matches. The Overall Judge splits 4. Winners are unaffected.
- **Air Tied** (J3 button/checkbox, FIS JH 6304.3.5.1) — the air comparison is tied, or neither competitor jumps. The six air votes are **withheld**: the air row pays **0 / 0** and the Overall Judge drops a point of scale. Note the deliberate handbook asymmetry — a speed tie awards the votes evenly (3 / 3) while an air tie withholds them (0 / 0). **Not allowed by default:** the meet's **Advanced** panel must enable "Air score tie allowed" or the button/ checkbox is hidden and the server refuses the submission — the Air Judge must pick a winner.
- If **both** speed and air are tied, the Overall Judge splits only **3** (their turns votes).

Every distributed total (25 / 25 / 19 / 19) is odd, so a tied match remains impossible. Whenever a tie declaration or NJ transition changes the Overall Judge's scale, their submitted split is automatically cleared and their tablet prompts a rescore on the new scale.

Aerials event setup

The aerials judging model added in v1.18.00 (called "v2" internally) is per-judge-per-jump: every scoring judge enters Air, Form, and Landing for every jump. Per FIS Judging Handbook 6004 / USSS 4110.

> Aerials events created **before** v1.18.00 use the legacy "component-specific judge roles" model and are read-only for historical results. New aerials events all use v2.

Picking the right Event Type

The Add Event modal's Category dropdown for aerials lists five Event Types. Each one drives the panel-size range and the HJ-may-score rule.

Event Type	Panel Size	HJ may score?
FIS OWG/WSC/WC	5–7	No
FIS NAC/NorAm	5–7	No
FIS Other	5 (locked)	No
USA National	2–5	No
USA Regional (default)	2–5	Yes

Pick the one matching your sanction document.

Panel size

After saving the event, set **Aerials Panel Size** on the Setup tab. This is the number of scoring judges (does not include the HJ unless the event type is USA Regional with HJ-may-score enabled).

- **5+ judges (FIS standard):** drop the high and the low per component independently, then sum the kept three (or middle four/five for 6/7-judge panels). Reduction method is fixed.
- **2–4 judges (USA reduced panels):** the **Reduction Method** dropdown becomes available. Pick one:
- **Sum All** (default) — no drops, sum every judge's score.
- **Drop High** — drop the highest, sum the rest.
- **Drop Low** — drop the lowest, sum the rest.
- **Average** — sum and divide by judge count.

The reduction method prints on the calculation report so the TD can verify.

Per-jump scoring

Each scoring judge submits per jump:

- **Air** — 0.0 to 2.0
- **Form** — 0.0 to 5.0
- **Landing** — 0.0 to 3.0

Per jump:

```
total_judges_score  =  sumKept(Air) + sumKept(Form) + sumKept(Landing)
jump_score          =  floor(total_judges_score × DD, 2dp)
event_total         =  sum across jumps
```

DD comes from the USSS Appendix C aerials chart (see Jump codes & DDs).

Seeding the panel

Once the panel size is set, go to the **Setup** tab and click **Seed N-Judge Aerials Panel**. This wipes any existing aerials judges and creates `N` rows with role `AeJudge1..N`, fresh short codes, and `judge_number = 1..N`. Then generate tablet URLs from the **Links** tab — see Seeding an aerials panel.

Tie-break

Per USSS 4110.4.3: Total › Air-no-DD › Form › Landing.

Why "v2"

Pre-v1.18.00 events used per-component judge roles (`AirJudge1`, `FormJudge1`, `LandingJudge1`) where each judge entered only one component. The v2 redesign brings the model into compliance with the rule books — every scoring judge evaluates all three components per jump.

Locking & unlocking events

Locking an event makes it **read-only** to officials and judges but leaves it fully visible on public scoreboards. Use this when an event is finalized and you want to prevent any accidental edits.

Effects of locking

When an event is locked:

- Every API endpoint that mutates the event data (score submissions, status changes, registration changes, judge assignments) returns HTTP 403.
- The event is **completely hidden** from the Officials Dashboard (and from the meet detail page if all events under it are locked).
- Judge tablets that try to score on a locked event get a "Locked" error.
- Public scoreboards and the broadcast overlay continue to display the final state.

Where to lock / unlock

Locks live on the event row (`events.locked` boolean column). Two places control them:

1. **Admin › Events** (Locking events from the Admin panel) — the canonical place. Per-event toggle plus Lock All / Unlock All per meet.
2. **Per-event** — via direct API in scripts or via the Admin panel.

There is no Lock button in the regular Officials interface — that's intentional. Locking is a privileged action.

What locking does not do

- It doesn't delete data — all rows stay in the database.
- It doesn't end the event for public viewers — they continue to see live scores and final results.
- It doesn't affect other events under the same meet — only the event you locked.
- It doesn't affect exports / PDFs / CSV — those still work normally.

Locking vs. hiding from Live Scores

Locking never removes an event from the public **Live Scores** page — spectators should still see a finalized event's results. If you want an event off the public listing (e.g., a **test event**), that's a separate, independent toggle: **Hide** on the Admin › Events page. An event can be locked, hidden, both, or neither.

Locking the whole meet

If every event under a meet is locked, the meet itself disappears from the Officials Dashboard list (which respects `excludeLocked=1`). Public surfaces still show the meet on `/livescores`. To bulk-lock all events of a meet, use **Lock All** in the Admin › Events page.

Unlocking

The same toggle. Unlocking is immediate — the event reappears in the Officials Dashboard and edit endpoints accept changes again.

Pre-v1.16.00 events

Lock support was added in v1.16.00. Events created before that version default to `locked=0` and behave as expected.

Why lock at end-of-meet

1. **Prevents accidental edits** during cleanup, archive, or post-meet review.
2. **Locks the audit trail** — no more rows are added to the audit log for that event.
3. **Keeps public surfaces clean** — the event continues to display its final state to spectators without risk of mid-screen flickers from a stray edit.
4. **Required for sanction archive** — the locked state can be exported and treated as the immutable record of the event.

Officials — Athletes & Registration

The Athletes database

StickIt keeps a **master Athletes table** that lives independently of any single meet. Every athlete you've ever registered, imported, or manually entered ends up here. Registering an athlete for an event creates a `registrations` row that points back to the master `athletes` row — so the same athlete appears identically across every meet they compete in.

Where to find it

The **Athletes** entry in the Officials sidebar opens the master list, paginated at 100 athletes per page. Columns:

- **First, Last**
- **Bib** (default bib, can be overridden per event)
- **USSA #** — the USSS membership number
- **FIS ID** (if any)
- **Club**
- **Div** — division code from the USSS People file (`SE`, `RM`, `IM`, etc.)
- **Birth Year**
- **Gender**
- **Edit** — inline-edit button

Soft-delete

Bulk deletes from the [Admin › Athletes management]() page are **soft-deletes**. The athlete row is not removed from the database — only the `deleted_at` timestamp is set. Effects:

- Soft-deleted athletes disappear from the master list, search, registration pickers, and CSV reconcile flows.
- Soft-deleted athletes are **not** deleted from past events. If they had registrations or runs in completed meets, those rows still display the athlete name and any historical scores.

Auto-restore

Re-adding a soft-deleted athlete via **Add from USSS Database** or via a SkiReg CSV import restores the existing row by clearing `deleted_at`. No duplicate is created. This is why a CSV import after a bulk delete just brings the right athletes back.

ALL CAPS auto-correct

Names entered as `SMITH` are automatically converted to `Smith` on save. This applies to every athlete create / update / sync path. Mixed-case names like `McDonald` or `O'Brien` are preserved as-is. The auto-correct handles:

- All-caps single names › name case (`SMITH` › `Smith`)
- All-caps multi-word names (`DE LA CRUZ` › `De La Cruz`)
- Hyphenated names (`SMITH-JONES` › `Smith-Jones`)
- Apostrophe names (`O'BRIEN` › `O'Brien`)

Mixed-case names are *not* changed. This is intentional — Mc/Mac, de la, von, etc. should be respected exactly as entered.

Division column

The Division column is sourced from the USSS People file's `D` field (Rocky Mountain, Eastern, Pacific Northwest, Intermountain, etc.). It's populated automatically during USSS sync, **Add from USSS Database**, and CSV reconcile flows. There is no manual override — to set or change a division, ensure the master USSS People file is current and re-sync.

Inline edit

Click the **Edit** pencil on any row. Editable fields: first name, last name, default bib, USSA#, FIS ID, club, birth year, gender. Save to update the master record. Past events are unaffected — past registrations carry the snapshot of the name and other fields at the time of registration.

When to use the master list

- Searching across meets for an athlete's history.
- Hand-editing typos that came in via CSV import.
- Manually entering an athlete who isn't in the USSS People file (a coach, an international guest).
- Auditing data quality before a critical meet.

Registering athletes for an event

Once an event is created, its **Registration** tab is where you list every athlete competing in that event. Each row links a master athletes record to this event, with a per-event bib number, run order, and status.

Five ways to add athletes

1. **From USSS Database** (fastest) — search the master USSS People file and add directly. The athlete is auto-created in your master Athletes table if not already there. See Importing from USSS People database.
2. **From SkiReg CSV** — bulk import the entire entry list from SkiReg. See Importing from SkiReg CSV.
3. **From Master Athletes Database** — search your local Athletes table.
4. **Manual Entry** — fill in the form by hand. See Manual athlete entry.
5. **Import Bibs from Event** (dual mogul only) — for dual events, copy bib numbers from the qualifier mogul event of the same gender (so bib 7 is bib 7 in both events).

Required fields

Every registered athlete must have:

- First name
- Last name
- USSS #
- Birth year
- Gender

A row missing any of these gets a **red row background** in the Registration table. Athletes with everything except a bib get a **yellow row background**.

> As of v1.16.31, the Run Order buttons and dual mogul seeding buttons are **not** blocked by missing bibs — you can build the run order first, then assign bibs in run-order order. They *are* blocked by missing names / USSS# / birth year.

Status

Each registration has a status: **Registered** (default) or **Scratched**. Scratched athletes are excluded from the run order and from results — but they remain in the table so you can un-scratch them if they show up.

The DNS / DNF / DSQ statuses you may have seen elsewhere are **run statuses**, not registration statuses. They're applied during scoring, not at the registration step. See Run statuses.

Bib numbers

Bibs can be assigned several ways — see Bib assignment. The bib is per-event: an athlete can have bib 7 in the mogul event and bib 14 in the aerials event of the same meet.

Run order

Once the registration list is built, expand the **Run Order** section to build the order athletes will compete in. See Building run order.

Collapsible sections

Both **Registered Athletes** and **Run Order** sections are collapsible (chevron toggle, default collapsed). They auto-expand when you add an athlete (any path) or when you modify the run order. The header bar with action buttons stays visible when collapsed.

Removing an athlete

Click the trash icon on an athlete's row. The registration is deleted; the athlete remains in your master Athletes database. If the athlete already has runs recorded, removing the registration will also remove those runs — confirmation modal appears first.

Importing from USSS People database

The **USSS People file** (a master roster of every USSS-licensed athlete, coach, and official) is the canonical source of truth for athlete data. Syncing it makes athlete entry instant and accurate — names, USSS#s, birth years, clubs, divisions, and FIS IDs all come from the file.

Syncing the USSS People file

The Officials sidebar's **USSS Database** entry opens a dedicated page (`/dashboard/usss`) with two buttons:

- **Sync Now** — fetches the current USSS People file from USSS servers and replaces the local copy.
- **Upload File** — manual upload of a `people.txt` file you downloaded yourself. Useful when the auto-sync URL changes or you have access to a pre-release file.

After sync, the page displays:

- Last sync timestamp
- List year + identifier (e.g. `2026 Fall`)
- Total record count + athletes count
- Source (`auto-sync` or `manual-upload`)

The sync is throttled — if the file hasn't changed since your last sync, the server returns `skipped: true` and shows "Already current". (Admins also have the same Sync Now / Upload File controls on the Admin › USSS People page.)

Add an athlete from the USSS database

Once synced, on the Registration tab of any event:

1. Click **+ Add Athlete › From USSS Database**.
2. A search modal opens. Type a name or USSS#.
3. Results filter live. Click an athlete to add them.
4. The athlete is created in your master Athletes table (if not already there) and registered for the event.

If the athlete was previously **soft-deleted** (Athletes database), this flow auto-restores them — no duplicate is created.

USSS People file structure

The file contains three record types:

- **C** — Competitor (licensed competing athlete)
- **CO** — Coach/Competitor (typically U15+ athletes with dual credentials — most are still primarily competitors)
- **O** — Official (judge, TD, course official)

The Admin › USSS People viewer (USSS People database (Admin viewer)) lets you browse the whole file, search by name / USSS# / club, and download as CSV.

When sync fails

If **Sync Now** fails with an error message:

1. Check the server's internet connection (can it reach the USSS server?).
2. Use **Upload File** as a fallback. Download the file from USSS's website directly and upload it via the modal.
3. Check the server's audit log for the underlying error message.

Automatic scheduled sync

A scheduled-sync job (`startScheduledSync` in `server/index.js`) runs nightly. It hits the same code path as the manual button, so any check-and-skip logic applies. Disable via `STICKIT_DISABLE_SCHEDULED_SYNC=1` if needed.

What gets synced into the master Athletes table

The USSS People file is read-only — it's a **lookup source**. Athletes are only created in your master Athletes table when you actually use them (via the registration **Add from USSS Database** flow, or via CSV import with name-fallback matching). This keeps the master Athletes table focused on athletes that are actually competing in your meets.

Importing from SkiReg CSV

SkiReg is the standard registration platform for USSS meets. After registration closes, the event director downloads an "Entry List" CSV from SkiReg. StickIt imports that file directly — no manual data entry, no Excel reformatting.

Steps

1. Download the entry list CSV from SkiReg.
2. On the event's **Registration** tab, click **+ Add Athlete › Import from SkiReg CSV**.
3. Pick the file.
4. A preview appears showing the count of athletes that will be added or updated. Bib conflicts (if any) are flagged.
5. Click **Confirm Import**. Athletes are added in bulk.
6. A blue banner appears asking **"Sync with USSS Database to fill in missing athlete data?"** — click **Yes** to enrich any partial records from the master USSS People file.

Expected CSV columns

SkiReg exports include these columns:

```
Last Name, First Name, Gender, Birth Year, USSS Member #,
Team, Bib, Category Entered, Quantity, Transaction Type,
Date of Birth, MerchSummary
```

Two known column-name variations are accepted: `Category Entered` and `Category Entered / Merchandise Ordered`. Both work.

How athletes are matched to your master Athletes table

Two-step matching:

1. **By USSS Member # first** (primary key).
2. **Fall back to case-insensitive name match** with birth year disambiguation. This prevents creating duplicate rows when an athlete was previously entered without their USSS#.

When a fallback match succeeds, the existing master record is updated with the SkiReg row's USSS#, birth year, gender, and club (filling in missing fields without overwriting existing ones).

Discipline matching

SkiReg's `Category Entered` field is loose ("Moguls", "Mens Mogul", "Mogul", "Dual Mogul", "Aerials"). The import logic:

- **Mogul event** — matches rows where Category contains `'mogul'` AND NOT `'dual'`
- **Dual Mogul event** — matches rows where Category contains `'mogul'` (any mogul row, since SkiReg often uses plain "Moguls" for dual entrants)

- **Aerials event** — matches rows where Category contains `'aerial'`

Dual mogul deduplication

For dual mogul events, SkiReg often emits two rows per athlete — one for "Moguls" and one for "Dual Moguls". Both rows match `Mogul` and `Dual Mogul` event imports. To prevent double-registration, the importer **dedupes by USSS#** within the import loop and merges bib numbers (whichever row has a bib wins).

Bib conflicts

SkiReg's bib field is sparse — only the mogul (single) row typically carries a bib; dual and banquet rows have empty `Bib`. The importer handles three modes:

- **Random** — assigns fresh bibs to anyone missing one, starting from 1, skipping taken bibs.
- **Run Order** — refuses to assign bibs (positional bibs would conflict). Build the run order first, then assign bibs via Bib assignment.
- **Copy from Event** — fills missing bibs only from athletes that also exist in the source event.

What if an athlete isn't in the master Athletes table?

If the SkiReg row carries a USSS# that doesn't match any existing athletes row and the name doesn't match either, a new athletes row is created from the SkiReg data. After import, run the **Sync with USSS Database** prompt to back-fill missing fields (birth year, club, FIS ID, division) from the master USSS People file.

Manual athlete entry

When an athlete isn't in the USSS People file (a coach competing, an international guest, a last-minute addition with no USSS# yet), enter them manually.

Steps

1. On the event's **Registration** tab, click **+ Add Athlete › Manual Entry**.
2. An inline form appears with these fields: - **First name** *(required)* - **Last name** *(required)* - **USSS #** *(required for USSS-sanctioned events)* - **Birth year** *(required)* - **Gender** *(required)* - **Club** *(optional)* - **FIS ID** *(optional)* - **Bib** *(optional — can be assigned later via [Bib assignment](#))*
3. Click **Add**. The athlete is created in your master Athletes table and registered for the event in one step.

When to use manual entry

- The athlete is competing under a temporary USSS# that hasn't synced to the People file yet.
- The athlete is a guest from another federation (foreign juniors, IFSA / FIS-only athletes, etc.).
- You're scoring a non-sanctioned scrimmage and don't want to bother with USSS# at all.
- You're testing the system and need a test record fast.

Required-field validation

The form refuses to save if a required field is missing. The Run Order buttons in the parent Registration section are gated on every non-scratched athlete having all required fields — so an incomplete manual entry will block run-order generation until you finish it.

USSS# placeholders

If the athlete genuinely has no USSS# (e.g., a foreign athlete), you can use a placeholder like `0` or `FOREIGN-1`. Just be consistent within the meet. The TD will want this flagged in the report — use the **Notes** field on the registration row (if visible) to add context.

ALL CAPS auto-correct

The same name normalization that applies to USSS sync and CSV import applies here — `SMITH` is saved as `Smith`. See Athletes database for the full rules.

What if the athlete name already exists?

If the entered USSS# matches an existing master Athletes row, that row is reused — no duplicate is created. If the name matches an existing row (by last+first+birth year), the existing row is updated with any new fields you've entered.

If you genuinely want a separate athlete record for someone with the same name (e.g., a junior and senior with identical names), differentiate by birth year — the master Athletes table treats `Smith, John (2005)` and `Smith, John (1985)` as distinct rows.

Bib assignment

Each registered athlete needs a bib number per event. The **Assign Bibs** button in the **Registered Athletes** section header opens the bib-assignment modal with four modes.

The four modes

1. **Random** — shuffles bib numbers 1..N (where N = registration count) randomly across athletes.
2. **By Run Order** — assigns positional bibs by current run order (bib 1 = first run, bib 2 = second run, etc.). Run order must exist for this to work.
3. **Copy from Event** — copies bib numbers from another event in the same meet (matched by USSS#). Useful when you want bib 7 to be the same athlete in both the mogul and aerials events of the same weekend. **Dual mogul events have a separate "Import Bibs from Event" button** in the Registered Athletes header — same mechanics.
4. **Fill In Missing** — keeps existing bibs and only assigns to athletes who don't have one yet.

Conflict handling

If you pick a mode that would overwrite existing bibs (Random, By Run Order), a confirmation modal shows:

- The count of bibs that will be overwritten.
- A button **Confirm Overwrite** or **Cancel**.

If no overwrites are needed, the modal applies immediately.

"Fill In Missing" rules

This mode behaves differently per parent mode:

Parent mode	Fill In Missing behavior
Random	Assigns sequential bibs 1..N to anyone without a bib, skipping bibs already taken.
Run Order	**Errors** — positional bibs would conflict with the existing assignment. Cancel and pick another mode.
Copy from Event	Assigns bibs only from athletes that also exist in the source event by USSS#.

When to assign

- **Before the run order** is built — random or copy-from-event are fine. The run order then sorts by bib if you choose `By Bib` order.
- **After the run order** is built — By Run Order is the typical choice, so bib 1 = first competitor.
- **From the master Athletes default bib** — some clubs assign a season-wide default bib. The Athletes table has a default bib column; CSV imports respect it.

Dual mogul bib import

For dual mogul events specifically, the **Registered Athletes** card header shows an **Import Bibs from Event** button (in addition to **Assign Bibs**). This opens an inline dropdown of same-meet, same-gender events; the import preview shows how many bibs will be copied, how many athletes have no match (skipped), and how many existing bibs will be overwritten. Confirm to apply.

This exists for dual mogul because dual mogul athletes are typically a subset of the mogul athletes from the same meet, and you want the bib numbers to align.

Validation

Bibs must be **unique within an event** but can repeat across events. The server enforces uniqueness via a database constraint. If you try to manually set a duplicate bib via the inline-edit form, the save will fail with an error.

Building run order

The **Run Order** section of the Registration tab determines the order athletes ski in for Run 1 (and Q1 for qualifier/finals events). Run 2 always reverses the standings of Run 1, so building Run 1's order well is the only operator decision.

Four build modes

1. **Random Order** — shuffles all non-scratched athletes randomly.
2. **By Age Groups** — orders athletes by USSS age class (youngest first: U7 › U9 › U11 › U13 › U15 › U17 › U19 › Sr › Vet), randomized within each age group.
3. **Manual reorder** — click the **(up)**/**(down)** arrows on each athlete row to move them up or down one position.
4. **Save Order** — persists the current order to the database. Until you save, drags / shuffles are local only.

Required-field gate

Run Order buttons are disabled until every non-scratched athlete has: first name, last name, USSS#, birth year. An amber warning above the section shows the count of incomplete athletes. (Bibs are *not* required — you can build the run order first and assign bibs by run-order order afterward.)

Confirmations

- **Random Order** / **By Age Groups** — confirmation modal appears only if a saved order already exists. First-time builds skip the confirmation.
- **Save Order** — no confirmation; saves immediately.

Hidden by default

The Run Order table is **hidden** until you've clicked a build button or until a saved order is loaded from the database. The header bar with the buttons stays visible — only the table inside is collapsed.

Age Class column

Once you've used **By Age Groups**, an extra **Age Class** column appears showing each athlete's computed USSS age class. The age class is derived from `birth_year` and the meet's `event_date` per USSS season rules (July 1 start). This column is informational; bib assignment and other operations don't depend on it.

How Run 2 order is determined

You don't build Run 2 order manually — the server reverses Run 1 standings automatically per FIS ICR 4205. Lowest-scoring athlete from Run 1 starts Run 2; highest-scoring athlete goes last. DNS / DNF / DSQ athletes from Run 1 are placed at the bottom in the order they appeared in Run 1.

For qualifier-finals events, the Final phase's run order similarly reverses qualifier standings, but only for the athletes that made the cut.

Locking the order

Once the order is finalized, click **Lock Order** (Locking run order). This:

- Disables all order-changing buttons.
- Adds a placement dialog whenever you add a new athlete or un-scratch a previously-scratched one.

Unlocking restores normal editability.

Visual feedback

Saved run order rows show their position in the **Order** column. Unsaved local changes show position with an asterisk (`*`) until you click **Save Order**.

Run order with age groups

The **By Age Groups** run-order builder groups athletes by USSS age class, youngest first, randomized within each group. This is the standard build mode for Devo events and any other event where the TD wants U7s to ski before U9s before U11s, etc.

USSS age class rules

USSS defines age class by the athlete's birth year vs. the **competition season**. A season starts July 1 — so for the 2025-26 season (any event held July 1, 2025 through June 30, 2026), athletes born in 2020 are computed as 5 years old and fall into U7.

The classes:

Class	Age range
U7	6 and under
U9	7–8
U11	9–10
U13	11–12
U15	13–14
U17	15–16
U19	17–18
Sr	19–34
Vet	35 and above

Build order

`By Age Groups` orders youngest age class first:

1. All U7 athletes (randomized within the group)
2. All U9 athletes (randomized within the group)
3. ... and so on up to Vet

Age Class column

After clicking **By Age Groups**, an extra **Age Class** column appears in the run order table showing each athlete's computed class. The column persists for the rest of the session even if you re-shuffle to **Random Order**.

Age group transition banner on tablets

When a run completes and the *next* athlete is in a different age group, an amber pulsing banner appears on Judge, Head Judge, and Timekeeper tablets:

> U11 group complete — U13 up next

It auto-dismisses after 15 seconds or on tap. **Devo events only** show this banner — Comp Series, RQS-EQS, and FIS events suppress it (those events are typically single-class or single-day formats).

The banner is purely informational — it doesn't pause scoring or block anything. It exists so judges can re-set their mental model (e.g., U7s vs. U17s look very different at the same score).

What if birth_year is wrong?

The age class is recomputed every time the run-order builder runs. If you fix an athlete's birth year on the Registration tab and re-click **By Age Groups**, the new class is used. The transition banner on tablets also uses the live-server compute, so corrections take effect immediately on the next run.

Manual override

If the TD insists on a custom order across age groups (e.g., bib draw, FIS world cup style), use **Random Order** or hand-reorder via the **(up)**/**(down)** arrows. The Age Class column still appears for reference, but order is whatever you save.

Locking run order

After the run order is finalized — typically the morning of the meet — click **Lock Order** in the Run Order section header. This protects the order from accidental edits and adds explicit placement controls for late additions.

Effects of locking

When the order is locked:

- **Random Order**, **By Age Groups**, the **(up)**/**(down)** arrows, and **Save Order** are all disabled.
- A placement dialog appears whenever you add an athlete (search, USSS, CSV, manual, bulk) or change a scratched athlete's status back to **Registered**.
- The lock state persists across page refreshes (`events.order_locked` column).

Placement dialog

When adding or un-scratching an athlete with a locked run order, you must pick where they go:

- **Start of Run Order** — bib position 1, all other athletes shift down by 1.
- **End of Run Order** — appended at the bottom.
- **Random Position** — placed at a random index in the existing order; existing positions shift around them.

This forces a deliberate choice — no athlete silently lands at the wrong place.

Scratching with a locked order

When you scratch an athlete (status changes from Registered › Scratched) and the order is locked, the athlete's `run_order` is cleared automatically (and their `phase_run_order` entries if phases exist). The positions of the remaining athletes are *not* compressed — there's a gap where they used to be. This preserves the meaningful order: bib 5 is still in slot 5 even if bib 4 is now scratched.

Unlocking

Click **Lock Order** again (it toggles). Unlocking restores all the editing buttons; you can shuffle, rebuild, or hand-reorder. After your changes, click **Save Order** and re-lock if you want.

When to lock

- **End of Day 0 / start of Day 1 morning** — once you've shaken out late drops, locked is the safest state through the event.
- **Just before bib assignment** — locking the order before running **Bib Assignment › By Run Order** guarantees bib positions don't shift mid-process.
- **After a TD-supervised draw** — once the TD has approved the order on paper, lock it so nothing changes inadvertently.

When to keep it unlocked

- **During open registration** — the order will change as people arrive.
- **During seed building for a dual mogul event** — you may want to rebuild seeds multiple times.

Server-side enforcement

`events.order_locked` is a database column, so the lock state is honored by every API endpoint, not just the UI. Even if you bypass the UI, the server refuses to change `run_order` on a locked event without the placement dialog's explicit `placement` parameter.

Officials — Judges

Adding & assigning judges
Judge tablet URLs
Seeding an aerials panel

Adding & assigning judges

Each event needs judges configured before scoring can begin. Judges live in the `judges` table, scoped per-event. The same person can be a judge in multiple events — each event has its own judges row for them.

Adding a judge

1. Open the event detail page › **Setup** tab.
2. Click **+ Add Judge**.
3. Fill in: - **Name** — first + last, free text. Shown on the HJ tablet and in PDF reports. - **Role** — pick from the dropdown. The list is filtered to roles still needed (e.g. if you've already added TL1, TL1 is removed from the dropdown). - **PIN** *(optional)* — 4-6 digits used by the judge when signing in on their tablet.
4. Click **Add**.

Roles per discipline

Mogul (Comp Series default):

- `TL1`, `TL2`, `TL3` — T&L judges
- `Air1`, `Air2` — Air judges
- `HeadJudge` — Head Judge
- `Timekeeper` — Timekeeper

Mogul (Devo / RQS-EQS default):

- `TL1`, `TL2` — T&L judges
- `Air1` — Air judge
- `HeadJudge`, `Timekeeper`

Dual Mogul:

- `DualTurns1`, `DualTurns2` — Turns judges
- `DualAir` — Air judge
- `DualTime` — Time judge
- `DualOverall` — Overall judge
- `HeadJudge` — Head Judge (no timekeeper for dual; time judge is one of the 5-point split judges)

Aerials (v2):

- `AeJudge1` through `AeJudge<N>` where N = `aerials_panel_size`
- `HeadJudge`

Friendly role display

The tablets display judge roles using a friendlier mapping:

- `TL1` › `T&L Judge 1`

- `Air1` › `Air Judge 1`
- `AeJudge1` › `Aerials Judge 1`
- `DualTurns1` › `Dual Turns Judge 1`

Internal values remain `TL1` etc. for routing/permissions. Only the display label is friendlier.

Editing a judge

Edit a judge's **name** and **USSS ID** inline, right in the Assigned Judges table — click into the field, type, and save. Roles aren't editable after creation; to change a judge's role, remove and re-add them (changing a role mid-event would orphan partial submissions).

Removing a judge

Click the trash icon. Delete. If the judge has submitted scores, the rows in `judge_scores` are removed too — confirmation modal warns first.

After adding judges

Generate tablet URLs from the **Links** tab — see Judge tablet URLs. For aerials specifically, see Seeding an aerials panel.

Number of judges defaults

The Add Event modal sets `num_tl_judges`, `num_air_judges`, `num_jumps`, etc. based on the category. The Setup tab's `+ Add Judge` dropdown then shows only the unfilled roles. To change the judge count after creation, edit the event (Editing an event) before scoring begins.

Judge tablet URLs

Each judge gets a **unique URL** to open on their tablet. The URL encodes the event and the judge — there's no login screen; opening the URL on a fresh tablet pre-binds it to that role.

Where to find them

Open the event detail page › **Links** tab. You'll see a table with one row per configured role:

Role	URL	Copy
TL1	`/judge/<event-short>/<judge-short>`	
TL2	`/judge/<event-short>/<judge-short>`	
Air1	`/judge/<event-short>/<judge-short>`	
Head Judge	`/headjudge/<meet-short>/<event-short>`	
Timekeeper	`/timekeeper/<event-short>`	
Scoreboard (public)	`/scoreboard/<event-short>`	
Overlay (broadcast)	`/overlay/<event-short>`	

Per-judge short codes

The judge URL contains a per-judge `short_code` — a small random string the server generates when the judge is added. This means:

- Two T&L1 judges in different events have **different** URLs.
- If you delete and re-add a judge, the new short code is different — the old URL stops working.
- The short code can't be guessed, so a judge tablet URL is effectively a credential. Keep them out of public-facing channels.

Aerials per-judge URLs

For aerials v2 events, each `AeJudge<N>` row has its own short code:

```
/aerials-judge/<event-short>/<judgeN-short>
```

This is different from the legacy aerials URL `/aerials-judge/<event-short>` (which still works as a fallback for pre-v1.18.00 events). The per-judge URL is what new events use.

Copy / open

- Click the **Copy** button to copy the URL to your clipboard. Paste into Slack, AirDrop, or email to the judge.
- Click the URL itself to open it in a new browser tab — useful for testing.

Tablet hardware

The tablets need to support a modern browser (iPad with Safari is the typical setup). Bookmark the URL on the home screen so the judge can re-open with a tap. Tablets should be set to "Guided Access" or kiosk mode to prevent accidental navigation.

What if a URL stops working

Three causes:

1. **Event is locked** — the URL returns "Event is locked" and refuses to load. Unlock the event from [Admin events](#).
2. **Judge was deleted** — the short code is no longer valid. Re-add the judge from the Setup tab to generate a new URL.
3. **Server-side short code was rotated** — rare, but possible if the database was restored from a backup. Open the **Links** tab and copy the new URL.

Print all URLs

There is no PDF "judge sheet" yet. To distribute the URLs, copy each one and paste into your shared Slack channel, AirDrop them to each iPad individually, or use the QR-code feature (if implemented in your version). A printable URL sheet is on the future roadmap.

Seeding an aerials panel

Aerials v2 events use numbered judge roles (`AeJudge1` through `AeJudge<N>`). Instead of adding judges one by one, **Seed N-Judge Aerials Panel** wipes existing aerials judges and creates the right number of slots in one click.

When to use

- New aerials event setup — fastest way to create the right panel.
- Changing the panel size mid-setup — wipes and rebuilds. Use before scoring starts.
- Recovering from a botched judge configuration — atomic reset.

Steps

1. Confirm **Aerials Panel Size** is set correctly on the Setup tab (e.g., `5` for a standard FIS panel, `3` for a USA Regional sub-panel).
2. Click **Seed N-Judge Aerials Panel** in the Setup tab's judge section.
3. A confirmation modal warns that existing aerials judges will be deleted. Click **Confirm**.
4. The server runs `POST /events/:id/judges/seed-aerials`: - Deletes all rows in `judges` where `event_id = ?` and role like `AeJudge%` or legacy aerials roles. - Inserts N new rows with role `AeJudge1` ... `AeJudge<N>`, fresh short codes, and `judge_number = 1..N`.
5. Names are left blank. Click each judge row to enter the judge's name (and optional PIN).

What it does not do

- Doesn't delete the HeadJudge row (kept).
- Doesn't delete any registrations, runs, scores, or other event data.
- Doesn't notify any tablets — if a judge tablet was open on an old judge's URL, that URL now 404s (or shows "Judge not found"). Re-share the new URLs from the Links tab.

After seeding

1. Fill in each judge's name on the Setup tab.
2. Generate tablet URLs from the **Links** tab.
3. Distribute the URLs.

Editing post-seed

Once seeded, you can:

- Edit any judge's name or PIN.
- **Cannot easily change the panel size** without re-seeding. Re-seeding is destructive to existing aerials judges (deletes & recreates), so do it before any scores are submitted.

HJ may score?

If the event type is **USA Regional** with HJ-may-score enabled, the HJ counts as one of the N panel judges. The panel size already accounts for this — for a USA Regional with panel_size=3 and HJ-may-score, you've got 3 scoring slots, one of which is the HJ. The other two are seeded as `AeJudge1` and `AeJudge2`.

Legacy aerials events

Pre-v1.18.00 events use component-specific roles (`AirJudge1`, `FormJudge1`, `LandingJudge1`) instead of `AeJudge<N>`. There is no seed button for those events. They render historical results unchanged but should not be used for new aerials competitions.

Reduction method check

After seeding, double-check that **Aerials Reduction Method** is set if the panel has fewer than 5 judges. For 2–4 judges, the options are: `Sum All` (default), `Drop High`, `Drop Low`, `Average`. Panels of 5+ use the FIS drop-high-and-drop-low rule automatically — no selection needed.

Officials — Scoring

Live scoring flow

Manual score entry

Voice manual score entry

Editing a finalized score

Run statuses: DNS, DNF, DSQ, NT

Head Judge review & approval

Live scoring flow

This is the end-to-end loop that runs all day at a live competition. Once the setup is done (event, judges, athletes, run order), the scoring tab is where you spend the day.

The loop, per athlete

1. **Start Run** — on the Scoring tab, click the blue **Start Run** button next to the athlete listed as `Up Next`. A confirm dialog appears (some versions). On confirm, the server creates a `runs` row with `status='scoring'` and broadcasts `run_started` to every connected tablet and public scoreboard.
2. **Judge tablets receive the bib** — each judge's tablet shows the new athlete's bib, name, and any prior runs. Judges enter scores and submit.
3. **Timekeeper enters the time** — when the athlete crosses the finish line, the timekeeper enters the run time (or NT for No Time).
4. **HJ tablet shows complete score** — once all judges and the timekeeper have submitted, the Head Judge tablet shows the full computed score with per-judge breakdown.
5. **HJ approves** — clicks **Approve & Submit**. The server finalizes the run, computes the new ranking, and broadcasts `score_update` to every viewer (officials, public scoreboard, overlay).
6. **Scoring tab auto-advances** — the next athlete in the run order becomes `Up Next`.

The loop repeats until every athlete in the phase has been scored.

Auto-finalize without HJ

If the event has no `HeadJudge` row in the judges table, the server **auto-finalizes** the run as soon as all required judge submissions are in. The HJ review step is skipped. This is the typical setup for a one-person scoring operation (Devo events especially) where the chief of score is both judge and head judge.

Auto-finalize after time

For events with a HJ but where the time judge submits last, the HJ tablet shows the complete score the moment time is submitted. There's a small delay (300ms) before showing the Approve & Submit button to give the HJ a chance to register what just happened.

Currently Scoring panel

The Scoring tab shows a **Currently Scoring** panel for any in-progress run:

- Athlete name + bib
- Each judge's submission status (Pending / Submitted)
- Time entry status
- Manual Entry button — see [Manual score entry]
- DNS button (yellow) — confirm-and-submit DNS for the current athlete

Up Next list

Below Currently Scoring is **Up Next** — the next 3–5 athletes in run order. The first row has a **Start Run** button + DNS button + Manual Entry button. Subsequent rows are reference only.

What can go wrong mid-loop

- **A judge submits a wrong score** — the HJ can reject just that judge's submission. See Head Judge review & approval.
- **A judge's tablet loses connection** — the judge can re-submit when it comes back; the polling fallback catches up.
- **The timekeeper enters the wrong time** — the HJ can reject the time submission, prompting a re-entry.
- **The Start Run was for the wrong athlete** — End Run from the HJ tablet or the Scoring tab, then Start Run on the correct athlete.

Phases and Run 2

For Best of 2 or qualifier/finals events, Run 1 / Q1 is the first pass. Once Run 1 is fully complete:

- Run 2 / F1 starts with a reversed run order (lowest from Run 1 first).
- The Scoring tab automatically detects the new phase and switches to the new run order.

For multi-phase events the **Phases** tab shows progress (which phase is in progress, which are finalized, which are sent for HJ review).

End of event

Once the last athlete is approved, the HJ tablet shows the **Final Review** screen with a green **Finalize Event** button. Clicking it sets `events.status='complete'` and broadcasts `event_finalized`. Public scoreboards show the final standings; the event disappears from the LIVE NOW strip on `/livescores`.

Manual score entry

The standard scoring path is tablet-based: judges enter scores on their tablets, the HJ approves, the server finalizes. But sometimes the tablet flow can't be used — a tablet is offline, scores are coming in over the radio from a remote judge, or you're recording a paper-scored event after the fact. **Manual Score Entry** is the fallback.

When to use it

- A judge tablet is offline or broken.
- Scores are being relayed by radio and you're typing them in centrally.
- You're recording a historical event from a paper score sheet.
- A score was missed by the tablet flow and you need to add it after the fact.

Where to access it

Several entry points:

- **Scoring tab › top of page › Voice Manual Entry** — opens the voice modal in paper-entry batch mode (mogul only; voice is hidden on aerials and dual mogul). See Voice manual score entry.
- **Scoring tab › "Currently Scoring" panel › Manual Entry button** — opens the keyboard modal for the currently-scoring athlete.
- **Scoring tab › "Up Next" row › Manual Entry button** — opens the keyboard modal for the first up-next athlete.
- **Results tab › per-row Manual Entry button** — opens the keyboard modal pre-populated with the existing scores for editing (this is also accessible via Editing a finalized score).

Standard mogul modal walkthrough

The modal lays out one section per scoring component:

1. **Time** — enter the run time in seconds (e.g. `27.34`). If the event has no speed (Devo), this section is hidden.
2. **Jump codes** — pick from the quick-select grid or the full dropdown for each jump.
3. **Per-judge T&L scores** — for each TL judge (TL1/TL2/TL3): - If component scoring is on: Carving, Absorption, Upper Body, Deduction. - If component scoring is off (RQS-EQS): Raw, Deduction.
4. **Per-judge Air scores** — for each Air judge (Air1/Air2): - Jump 1 score, Jump 2 score.

Each input has +/- buttons for fine tune. The running calculated total updates live at the bottom.

A **Submit** button persists the run. A **Cancel** button discards. A **Status Override** lets you submit DNS / DNF / DSQ instead of a numeric score — each asks for confirmation first, so a stray tap can't zero out an athlete.

Two guard rails on submit:

- **Implausible deductions** — a deduction above the full-fall maximum of 6.0 (almost always a score typed into the deduction field) triggers a confirmation. On confirm the value is accepted as entered — it is never refused or capped. The same soft stop exists on the judge tablets.

- **Finalized rounds** — if the run's round is already finalized, an amber notice appears in the modal and Submit asks for explicit confirmation ("…saving will change published rankings"). The edit stays allowed — it is the rescue path for a missed athlete — and every post-finalization edit is recorded in the audit log (`edit_after_finalization`).

Abandoning a stuck run

If a run gets stuck in scoring (started during an outage, judges never submitted, the wrong athlete was started), the **Currently Scoring** panel on the Scoring tab has an **Abandon Run** button. It deletes the run and all of its judge scores regardless of how many were submitted — a two-step confirm, audit-logged, and available only from the Officials UI (never on a tablet). The panel also warns when a run has been sitting for several minutes with no scores, so a stuck run is visible immediately. Note that while a run is in scoring, no new run can be started (tablet mode) and the round cannot be finalized — Abandon Run is the escape hatch.

Aerials manual entry

Aerials events (panel size set, per-judge-per-jump scoring) get their own manual entry modal: a grid of one column per scoring judge × one row group per jump, with Air / Form / Landing inputs mirroring the Head Judge tablet's layout. Jump-code pickers, range validation, and confirmed DNS/DNF/DSQ overrides are all built in. Editing an existing run pre-populates every judge's values. The total is computed by the exact same engine path as tablet scoring.

(Older releases refused aerials manual entry and required the per-judge tablets — that limitation was removed in v1.25.00.)

Dual mogul

Dual mogul has its own manual entry path — see Dual mogul setup. The **Manual Score Entry** button on the dual scoring tab opens a 5-judge entry modal for the current match.

Partial scores

The modal supports partial submissions. Leave a judge's score blank and the run is recorded as "incomplete" — the score appears with `–` placeholders. This is useful for capturing what you have so far when an event was running on paper and you're back-filling.

Voice option

In v1.20.00+ a **Voice Entry** button appears in the manual entry modal header. Clicking it closes the keyboard modal and opens the voice modal in tablet-edit mode with the athlete pre-selected, skipping the bib screen. See Voice manual score entry.

Voice manual score entry

A hands-free, voice-driven workflow for the chief-of-score role. Built on Deepgram Nova-3 streaming speech recognition. Voice is purely *additive* to the existing scoring paths — all formulas and routes are unchanged, and everything voice can do can also be done with the keyboard.

When to use it

- You're scoring paper-mode all day and want to keep your hands on the paper sheets while your voice drives the modal.
- You have a Philips SpeechMike or Nuance PowerMic whose buttons are mapped to the hotkeys below.
- The keyboard manual entry is too slow for the run cadence.

Where to access it

- **Scoring tab › Voice Manual Entry button** — batch (paper) mode. Bib confirmation › guided dictation › review › submit › next-bib prompt › repeat. A styled **Close** button on the bib screen exits at any natural break.
- **Manual Score Modal › Voice Entry header button** — tablet-edit mode. Skips the bib screen with the athlete pre-selected.

Voice is hidden on **dual mogul** (bracket scoring is fundamentally different) and on **all aerials events** (aerials uses its own per-judge entry paths).

The wizard model

The dictation screen is a **guided wizard**, not free-form. A blue-glow cursor highlights one field at a time. The operator speaks only the value for that field. Voice commands:

Command	Action
"Next"	Advance the cursor to the next *blank* field
"Redo"	Clear the value of the active field; cursor stays
"Submit" / **"Done"**	Stop the mic and jump to the Review screen
"No Time" / **"NT"**	Only on the Time field — marks the run No Time

The stay-put rule

Saying "six point five" while the cursor is on Carving lands `6.5` in Carving — and the cursor **stays on Carving**. Saying another number replaces the value (self-correct by simply re-speaking). You must explicitly say "Next" (or press a Next key) to move on. This eliminates false advances on partial speech-recognition returns.

Hardware keys (SpeechMike / PowerMic / keyboard)

All field navigation can be driven from hardware buttons instead of voice. macOS intercepts the bare F-keys at the OS level, so each action also has a **Ctrl+key alias** that always works:

Action	Key	macOS-safe alias
Previous field (Dictation) / Re-record (Review)	`F1`	`Ctrl+[`
Start / stop the microphone	`F2`	`Ctrl+,`
Next field; on last field › Review. On Review › Submit	`F3`	`Ctrl+]`
Next *blank* field; on last field › Review	`F4`	`Ctrl+\`

The mic toggle (`F2` / `Ctrl+,`) also works on the **bib screen**, so a hardware Record button drives the whole batch flow.

SpeechMike setup: in Philips Device Control Center, open your Custom profile and assign the device buttons to the Ctrl+key combos above using the Hotkey option. Suggested mapping: Rewind ‹‹ › F1 action, Record • › F2 action, Forward ›› › F3 action, EOL › F4 action, Play › disabled. PowerMic users configure the same hotkeys in the Nuance driver utility. AirPods / USB headsets work with the on-screen Start/Stop button.

Click-to-edit-target

Tap any field in the script during recording to set it as a temporary edit target (dashed amber ring). The next spoken value lands in the clicked field — not the wizard cursor's position. After the value lands, the edit target clears and the wizard cursor resumes where it was. Two-cursor model: wizard position is sequential; click is random-access.

Re-entering an athlete who already ran

On the bib screen, speaking or typing a bib that already has a run in the current round shows a warning: *"Bib X has already run in this round. Re-entering will OVERWRITE their existing scores. Continue?"* Confirm to proceed — the wizard runs normally and submits through the score-edit path against the existing run. Decline and the bib is rejected with a pointer to **Edit Score** on the Scoring tab. Status overrides (DNS/DNF/DSQ) on an already-run athlete route the same way.

Status overrides — buttons only

Three buttons in the modal header — **DNS / DNF / DSQ** — each open a confirm dialog and submit the run with that status, overriding any partial scores. **Voice intentionally does NOT recognize DNS/DNF/DSQ as commands** — the toolbar is the only path, so a chance mishearing can never zero out an athlete's scores.

Jump code validation

When the modal opens it fetches the actual jump-code list for the event's discipline + gender. Every parsed jump code is validated against that set. Codes not in the table flash red (`[no]`) with the recognized text shown so you can immediately click + re-record. Submit is blocked while any jump code is invalid.

Speaking grabs — basic vs advanced (v1.26.00)

Since the FS-13 rule change, basic (`g`) and advanced (`G`) grabs are different jumps, and the voice vocabulary distinguishes them:

- Saying **"grab"** always means the **basic** grab: "back grab" › `bg`, "three grab" › `3g`, "seven o grab" › `7og`.

- Say **"big grab"** or **"advanced grab"** for the **advanced** grab: "back big grab" › `bG`, "three advanced grab" › `3G`.
- Letter-by-letter dictation ("b g") resolves to the basic (lowercase) code; use the "big grab" phrases to get the advanced code.

Case is preserved through parsing, validation, and the Review screen — a typed correction of `bg` to `bG` on the Review screen changes the jump.

Confirmation Review

After "Submit" / "Done", a two-column Review screen shows the raw transcript on the left and parsed values on the right as editable inputs. Each row has a status glyph: green [yes] (ok), amber `!` (out-of-range), red [no] (invalid jump code), or gray `·` (empty). **Typed corrections get the same range checks as spoken values** — an out-of-range typo flags amber just like a mishearing. Submit is gated on every required field being filled AND every jump code being valid. Re-record (`F1`) discards everything and re-enters the wizard.

Cancelling

Closing the dictation screen with values already entered asks *"Discard everything entered for this athlete?"* first, so a stray tap can't lose a half-dictated run.

Failure modes

- **Voice service unavailable / network outage** › modal shows "Voice service unavailable. Use keyboard manual entry." Close and use the keyboard path; no partial data is written.
- **Microphone permission denied** › clear error message in the modal.
- **Bad parse / invalid jump code** › field shows red. Click the field to retarget and re-record, or type-correct on the Review screen.

Server requirement

`DEEPGRAM_API_KEY` must be set in the server's environment. Without it, the voice modal renders the "Voice service unavailable" error (the Admin Dashboard shows whether the key is configured). Audio is never persisted — only the resulting transcript text is stored with the run for post-event audit.

Editing a finalized score

Sometimes a finalized run needs to be corrected — a judge transposed a digit, the wrong jump code was recorded, the time was misread. StickIt supports score editing through the same Manual Score Entry modal pre-populated with the existing values.

Steps

1. Open the event › **Results** tab (or the **Scoring** tab's score history).
2. Find the run you want to edit.
3. Click the **Manual Entry** / **Edit** button on that row.
4. The Manual Score Entry modal opens with every existing value pre-filled.
5. Change the fields you need.
6. Click **Submit**.

The server replaces all `judge_scores` for that run with the new values, recomputes `total_score` / `turns_score` / `air_score` / `speed_score` / `air_score_no_dd`, and broadcasts `score_update` to every viewer.

Best-run selection

For Best of 2 events, editing one run may change which run is the athlete's best. The server re-runs `pickBestRun` (with the FIS-compliant tie-break) after the edit — so rankings update automatically. The Best Score column on the Results tab reflects the new best.

Tie-break re-application

If the edit pushes the athlete's score across a tie with another athlete, ranks shift accordingly. The `tieBreakMogul` / `tieBreakAerials` comparator runs at every results-read site, so the new rank is computed correctly. There's nothing to manually re-run.

Dual mogul edit

For dual mogul, completed matches show an **Edit Scores** button regardless of `score_entry_mode` (as of v1.16.17). Clicking it opens the paper-score modal pre-populated with the existing 5 judge points. Editing a non-leaf match's winner does **not** auto-recompute downstream matches — if the change inverts who advanced, you'll need to manually re-do later rounds.

When editing is allowed

- **Any time** before the event is finalized — no restriction.
- **After Best of 2 / qualifier finalization** but before event-level finalization — still allowed; the affected phase is automatically updated.

- **After event finalization (`events.status='complete'`)** — the **Edit Scores** button is **hidden** for dual mogul (v1.19.01); for standard mogul it's still available but be careful. A future "re-open finalized event" admin action will provide a cleaner path.

Audit trail

Every edit is logged in `audit_log` with the user, timestamp, run ID, old values, and new values. View on the Audit log page.

What the public sees

Public scoreboards refresh within 5 seconds of the edit (via the WebSocket `score_update` broadcast). The athlete's new score appears immediately. There's no "edited" badge — the new score replaces the old one cleanly.

Voice-edit shortcut

In v1.20.00+, the Manual Score Entry modal's header has a **Voice Entry** button. Clicking it closes the keyboard modal and opens the voice modal in tablet-edit mode with the athlete pre-selected. See Voice manual score entry. Voice edit follows the same wizard model as voice paper-entry.

If only the time is wrong

For a fix limited to the run time, the keyboard modal is the fastest — just open it, change the Time field, and submit. No need to touch any of the per-judge fields. The server only updates what changed.

Run statuses: DNS, DNF, DSQ, NT

Beyond a numeric score, every run can have a **status** indicating something didn't go to plan. Statuses are stored on the `runs` row and trigger specific display and ranking behavior.

The statuses

Code	Means	Counted in CC?	FFSP	Results placement
DNS	Did Not Start	No	0	Bottom of its tier, after DNF
DNF	Did Not Finish	Yes	Formula or 0 (1st round)	Bottom of its tier, first among statused
DSQ	Disqualified	No	0	Absolute bottom of the entire event
NT	No Time (mogul only)	N/A	N/A	Speed score = 0 (run ranks normally)

CC = Counting Competitors (used for FFSP — see FFSP points (dual mogul)).

RNS (Refused to be Scored) was retired in v1.26.00. It was a legacy Winfree convention with no equivalent in the USSS Comp Guide or FIS ICR. It can no longer be assigned, but historical events that contain RNS runs still display them everywhere; a legacy RNS orders between DNF and DNS in results and is transmitted to USSS as DNF.

Where to set each

- **DNS** — yellow "DNS" button on the Up Next athlete row (Scoring tab, HJ tablet, Timekeeper tablet). Confirmation modal appears. Or the Manual Entry modal's "Status Override" picker.
- **DNF** — HJ tablet has a DNF button. Used when the athlete started but didn't finish (fell, missed gate, equipment failure). Confirmation required. **Gate fault is formally a DNF** (Spring 2026 FIS rule), and a competitor who intentionally re-enters the course after any DNF receives a **DSQ**.
- **DSQ** — HJ tablet has a DSQ button. Used when the athlete is disqualified per TD ruling.
- **NT** — Timekeeper tablet has a red **No Time** button. Confirmation modal appears. Used when no valid time could be recorded but the athlete did finish.

NT specifics

NT produces a speed score of `0`. Internally stored as `run_time = -1` (sentinel). The scoring engine's `calcSpeedScore` returns 0 for non-positive times. **Finalization proceeds normally** — the run is not blocked waiting for time.

The Time column shows "NT" across all surfaces (Timekeeper, HJ, Scoreboard, EventDetail, PDFs, exports) — never `-1`.

How statused athletes are ordered (v1.26.00)

Statused athletes are no longer dumped unordered at the bottom. Per USSS 4012.3:

- **DNF and DNS are phase-scoped.** Each places the athlete at the **bottom of the tier (phase) in which the status occurred**, ahead of every athlete in lower tiers. Example: a DNS in a Final round of 8 places 8th — ahead of all Final 1 and Qualification athletes.
- **Within a tier** the order is: scored athletes (normal ranking), then DNF, then RNS (legacy data only), then DNS.
- **DSQ is event-scoped.** A DSQ places at the absolute bottom of the entire event, below every tier and every other statused athlete, regardless of which phase it happened in.
- **Statused athletes consume numeric places**, so lower tiers continue numbering after them. In a Final of 8 with 6 scored, 1 DNF, and 1 DNS: scored athletes take places 1–6, the DNF is 7th, the DNS is 8th, and the qualification tier begins at 9th. The status code (not the number) still prints in the Place column.
- **Ties among same-status athletes** in the same tier share the place and list by bib number ascending; the next place skips accordingly.
- **Multiple statused runs** (e.g. Best of 2 with DNF in run 1, DNS in run 2) resolve to a single status by precedence: DSQ, then DNF, then DNS. An athlete with at least one scored run in the tier ranks normally on it and is not flagged.
- **Qualifier/Finals tier assignment:** an athlete who scored in Qualification and then DNSed Final 1 places at the bottom of the Final 1 tier — not in the qualification tier on their qualification score.

This ordering is identical on the Results tab, the public Scoreboard, all results PDFs, and the USSS XML transmit.

DNS / DNF / DSQ display

The score column on every surface shows the status text instead of `–` for pending runs. Examples:

- Scoreboard: Total column shows `DNS` / `DNF` / `DSQ` instead of blank.
- Results tab: Total column shows status text.
- PDF reports: Total column shows status text.
- Individual phase columns: Show status text for that run only.

Component-level display

For DNS / DNF / DSQ athletes, only the **Total** column changes. Component breakdowns (Turns, Air, Speed, Time) stay blank or `–`. This keeps the row visually distinct without spurious zeros polluting the breakdown columns.

Re-opening a DNS / DNF / DSQ run

If an athlete was marked DNS but then actually starts (showed up after the bib draw), you can clear the status from the Scoring tab's run history:

1. Open the run's Manual Entry modal.
2. The Status Override picker shows `None` / `DNS` / `DNF` / `DSQ`. Pick `None`.
3. Enter the actual scores.
4. Submit.

Re-opening removes the status, replaces with a numeric score, and re-runs the standings.

Head Judge review & approval

The Head Judge tablet is the central review point in the live scoring flow. Once every required judge has submitted, the HJ tablet shows the full computed score and waits for explicit approval before finalizing.

What the HJ sees during a run

Top of screen:

- **AthleteBar** — bib, name, club, current run number / phase.
- **Status squares** — three indicators showing T&L Judges (`3/3`), Air Judges (`2/2`), Time (`[yes]`). Green when complete, gray when pending.
- **HC Mode toggle** — switch to high-contrast view.

Center of screen, during pending state:

- **Per-judge review rows** — one row per judge with score, jump codes (Air judges), and a small **Reject** button.

Bottom of screen:

- **Run Status grid** — three large buttons: DNS, DNF, DSQ. Each opens a confirmation dialog before submitting (see Run statuses). RNS was retired in v1.26.00; legacy values still display.

Once all judges have submitted

The center of the screen switches to:

- **Calculated score panel** — Turns, Air, Speed (or Time Points), Total. All values match exactly what will be persisted on approval.
- **Approve & Submit** button — green, large, bottom-of-screen.
- **Send Back to Scoring** button — amber, opens the run back to specific-judge rejection.

Rejecting a single judge's submission

Click the **Reject** button next to a single judge's row. A confirm dialog appears:

> Reject TL1's submission and ask for re-entry?

Confirm to clear that judge's score from the database. The judge's tablet sees the rejection on its next poll cycle (or via the live WebSocket message) and resets to the score-entry state with their previously-submitted value cleared. The HJ tablet returns to the pending state.

This is the surgical tool — used when one judge entered the wrong score but everyone else is fine.

Sending the whole run back

Less surgical. Useful when multiple judges need to re-enter, or when a fundamental error (wrong athlete bib showing on tablets) requires a full reset. Clicking **Send Back to Scoring** clears the runs row's HJ-pending status; the Scoring tab gains an "Edit Scores" affordance and the operator can adjust there.

Jump code mismatch (Air judges disagree)

If the two Air judges submit different jump codes for the same run (compared exactly, case included — `bG` vs `bg` is a mismatch), both judges' scores are still recorded (v2.3.00; earlier builds refused the second judge's submission outright). The HJ tablet's Air Judges card shows a red **Jump Code Mismatch** box with one line per Air judge (their two codes) and:

1. **Accept These Codes** next to either judge — the run's official codes and DDs become that judge's codes, both judges' air scores are scored against them, and both Air judge tablets show **Air Codes Reconciled by Head Judge**. The original codes are kept in the audit log (`air_codes_reconciled`).
2. **Reject Both Codes** — the same action as **Reject Codes**: nulls the run's codes, deletes both Air judges' scores, and returns both tablets to code entry so they re-coordinate and resubmit.

Finalize and Publish Score stays disabled (and the server refuses approval) until the mismatch is resolved, alongside the usual missing-scores gate.

Reject codes (no mismatch)

Reject Codes on the Air Judges card clears the run's codes and both Air judges' scores at any time — for example when both judges agreed on a wrong code.

Approving

Click **Approve & Submit**. The server:

1. Finalizes the run (`runs.status='complete'`).
2. Computes ranking.
3. Broadcasts `score_update` to officials, public scoreboards, and the overlay.
4. Advances the Scoring tab to the next athlete in run order.

End of phase

Once every athlete in the phase has been approved, the HJ tablet shows a phase summary card with **Approve Phase** / **Send Back to Scoring** buttons. Approving the phase locks it in `event_phases.status='finalized'`. Sending back re-opens any individual run for re-editing.

End of event

After the final phase is approved, the HJ tablet shows the **Final Review** screen — a combined results table with **Approve & Finalize Event** and **Send Back to Scoring** buttons. Approve to mark `events.status='complete'`; this is the final lock that ends the live scoring loop.

Dual mogul HJ flow

For dual mogul, the HJ tablet manages match-by-match approval. After every bracket match is complete, the HJ tablet shows the full bracket tree with **Approve & Finalize Event** and **Send Back to Scoring** buttons. See [Dual mogul setup].

Officials — Results & Reports

Reading the results table

The **Results** tab on the event detail page shows the live, sortable ranking for the event. It updates automatically as runs are finalized. The layout varies by event type (single run, Best of 2, qualifier/finals, dual mogul).

Standard mogul columns

Column	Meaning
Place	Rank (Olympic-style skip rank — see Tie-break rules)
Bib	Athlete's bib for this event
Athlete	`Last, First`
Club	Club abbreviation
Turns	Turns score component (max 60)
Air	Air score component (max 20, post-DD)
Time	Run time in seconds (or `NT` / `DNS` / `DNF` / `DSQ`)
Speed	Speed score component (max 20)
Total	Sum: Turns + Air + Speed (max 100)

Click any column header to sort. Default sort is by **Place** ascending.

Best of 2 columns

For Best of 2 events, an additional column structure:

Column	Meaning
Run 1	Total for Run 1
Run 2	Total for Run 2
Best	The higher of the two (the official score)
Place	Rank by Best

The starred run is the athlete's better run. If editing changes which run is better, the star moves automatically (FIS-compliant `pickBestRun` is applied at every read site).

Qualifier/Finals columns

Column	Meaning
Q1	Qualifier 1 total
F1	Final 1 total
F2	Final 2 total (if configured)
Place	Final rank — Olympic-style skip with tiered ranking (F2 > F1 > Q1)

Athletes who didn't make the cut to F1 are ranked by Q1 standings below those who did.

Aerials columns

Column	Meaning
Jump 1	Jump 1 score (per-judge sum × DD, floored to 2dp)
Jump 2	Jump 2 score
Air-no-DD	Raw air execution score (sum of Air component means per jump, no DD)
Form	Form component total
Landing	Landing component total
Total	Sum across jumps

For v2 aerials events. Legacy aerials events use a different column set.

Dual mogul

Dual mogul shows a **bracket view** instead of a flat table. See the [Reading a Scoreboard] topic for the spectator-facing version; the Results tab's bracket is essentially the same.

Status text in score columns

DNS / DNF / DSQ runs show their status text in the **Total** column (and in per-phase columns for multi-phase events). Component columns (Turns, Air, Speed, Time) remain blank for those rows. See [Run statuses].

Click-to-expand

For mogul events, clicking an athlete row reveals a per-run detail row showing per-judge breakdowns:

```
Run 1: TL1 5.5  TL2 4.5  TL3 1.2  S/bp  A1 3.4  A2 4.5  T: 23.23  TPt: 16.3  Total: 56.23
```

Click another row to switch; click the same row to collapse. (This is the same UX as the public scoreboard's expand.)

Live updates

The Results tab listens for WebSocket `score_update` messages and re-fetches the standings on every new finalized run. There's no need to refresh the browser.

Printing

Use the PDF buttons in the Results tab header to generate the Event Results Summary PDF — see [PDF reports].

Tie-break rules

When two athletes have the same total score, StickIt applies a strict tie-break ordering per the FIS / USSS rule books. The tie-break is applied at every read site (Results tab, Scoreboard, PDFs, CSV exports, USSS transmit XML) so all surfaces stay consistent.

Mogul — FIS ICR 4207.3

1. **Total** — higher total wins.
2. **Turns** — higher turns score wins.
3. **Air-no-DD** — higher raw air execution score wins. (This is the per-judge air mean per jump, summed, with no DD multiplication — stored on `runs.air_score_no_dd`.)
4. **Speed** — higher speed score wins.

All comparisons use 0.001 epsilon to handle floating-point drift.

Aerials — USSS 4110.4.3

1. **Total**
2. **Air-no-DD**
3. **Form**
4. **Landing**

Form and Landing are summed across jumps. For events created at or after v1.18.00 (v2 model), per-jump Form/Landing are available; for legacy events, the single Form/Landing per run is used.

Dual mogul

Dual mogul uses head-to-head match wins, not score totals. Ties within a 5-judge match are resolved by individual judge components (Turns, Air, Time, Overall) per FIS dual mogul rules. There's no "total" tie to break at the event level.

Olympic-style skip rank

Per FIS ICR 4207.3.4: when athletes tie at the unbreakable level (every step of the tie-break sequence is equal), they **share a rank**. The next rank then skips by the size of the tied group:

Place	Athlete	Total	Turns	Air-no-DD	Speed
1	Alice	78.45	55.0	12.5	11.0
2	Bob	76.20	52.0	12.0	12.20
3	Carol	75.10	50.0	13.0	12.10
3	Daniel	75.10	50.0	13.0	12.10
5	Edward	74.00	49.0	12.5	12.50

Carol and Daniel tie at every level. Both get rank 3. Edward is rank **5**, not rank 4 — the rank skips to account for the two-way tie.

Cut-line tie expansion

When a tied rank straddles a cut line (e.g., qualifier › finals cut at 16 athletes, but ranks 16 and 17 are tied), **all** tied athletes advance. The field expands as needed. See [Multi-phase events] for details on how this affects phase creation.

Best-run selection (Best of 2)

For Best of 2, if both runs total to the same score, StickIt picks the FIS-stronger run via the same tie-break sequence (Total › Turns › Air-no-DD › Speed). This affects which run is starred as the athlete's best in the Results tab and which run's component scores are used for the athlete's ranking.

Persisting air_score_no_dd

The pre-DD raw air execution score is computed automatically when each run is finalized:

```
air_score_no_dd = avg(per-judge air for jump 1) + avg(per-judge air for jump 2)
```

floored to 2dp. Single-jump events double the one jump's average.

A startup backfill (`backfillAirScoreNoDd`) populates the column for any pre-v1.16.23 runs that don't have it. Manually-entered runs missing per-judge data fall back to `air_score` (post-DD) for the comparison only.

Why this matters

A poorly-implemented tie-break can swap places at the top of a podium. StickIt's tie-break runs at every results read site, not just the final PDF generation — so live scoreboards and broadcast overlays show the same order the TD will see in the official results.

FFSP points (dual mogul)

FFSP (Freestyle Skiing Points) is the USSS point system that determines national / divisional rankings across the season. StickIt computes FFSP automatically for **dual mogul events only** as of v1.16.19 — standard mogul and aerials FFSP are not in scope this release.

When points are computed

- Only after the event is finalized (`events.status='complete'`).
- Pre-completion the Points column on the dual scoreboard's **Place** tab shows `—`.

Tier selection (auto, no UI config)

The tier is picked automatically based on the event's flags and the number of Counting Competitors (CC):

Tier	1st	2nd	3rd	Condition
Divisional Championships	1000	970	950	`events.is_divisional = 1`
Divisional Events (CC >= 15)	900	875	855	`is_divisional = 0` and CC >= 15
Divisional Events (sub-15)	800	775	760	`is_divisional = 0` and CC < 15

Counting Competitors (CC)

CC = total placements **minus** DNS / DSQ / scratched athletes.

- **DNFs count toward CC** — they participated, even though they didn't finish.
- DNS, DSQ, and scratched athletes do not count.

Formula for 4th place onward

```
ffsp(rank) = max(0, tier3rd - (rank - 3) × (tier3rd / CC))
```

So at a Divisional Events (sub-15) event with CC = 12 and `tier3rd = 760`:

- 4th place: `760 - 1 × (760 / 12)` = `760 - 63.33` = **696.67**
- 5th place: `760 - 2 × 63.33` = **633.33**
- ... linear decay down to 0 at rank `CC + 3`.

Special cases

- **DNS / DSQ / scratched** › 0 FFSP, excluded from CC.
- **First-round DNF** — an athlete who DNFs in the literal largest-numbered bracket round (e.g., Round of 32) gets **0 FFSP**.
- **Advancing-via-bye DNF** — an athlete who advanced via a bye and then DNFs in their first played match still receives formula-based FFSP.

Where it shows

- **Public Scoreboard › Place tab** — Points column.
- **Final Place List PDF** (`POST /api/pdf/dual-results`) — Points column.
- **CSV / Excel exports** — Points column for dual mogul events.

Helper function

The pure helper `computeDualFfsp({ event, bracket, placements })` in `server/dual/ffsp.js` does all the math. Both the results endpoint and the PDF route call it, so the values are guaranteed identical. No DB writes — FFSP is recomputed on demand so score edits trigger automatic recompute.

Why dual mogul only

Standard mogul and aerials FFSP tables aren't yet built into StickIt. They use a different ladder (FIS race points, with course difficulty multipliers and field penalties). For now, those scores must be computed externally — typically by hand-typing the rank list into the USSS FFSP calculator spreadsheet.

PDF reports

StickIt generates a family of PDF reports server-side via pdfkit. They're available from the **Reports** tab on the event detail page (or via direct API calls from `server/routes/pdf.js`).

Event Results Summary

The headline deliverable. Lists every athlete in rank order with all component scores. Format adapts by event type:

- **Standard mogul** — Place, Bib, Athlete, Club, Turns, Air, Time, Speed, Total.
- **Best of 2** — adds Run 1 / Run 2 columns; Total = the better run, with the starred run column indicating which one.
- **Qualifier/Finals** — adds Q1 / F1 / F2 columns.
- **Aerials** — Place, Bib, Athlete, Club, Jump 1, Jump 2, Air-no-DD, Form, Landing, Total.
- **Dual mogul** — uses the Final Place List format (see below).

Includes meet header (name, location, date), event header (name, discipline, gender), and tie-break notation.

Check Sheet by Bib

Sorted by bib number. Used by judges and the TD during the meet for quick lookup. One row per athlete with each run's score on its own row.

Check Sheet by Run Order

Same data, sorted by run order. Mirrors how athletes appeared on the start list.

Start List

Run order, bibs, names, clubs. Used pre-event to confirm the day's lineup.

TD Report (USSS Freestyle Technical Delegate Report)

The official sign-off document. Lists the meet, every event, results summary, judging panel, and a "Send to" line:

> USSS Freestyle Head TD, ResultPackets@ussa.org, Organizing Committee, Division Head TD

Updated in v1.16.03 — title and instructions reference "USSS" (not "USSA").

Dual Bracket (compact)

Tree-format bracket PDF. Page count scales by bracket size:

- **16 athletes** › 1 page
- **32 athletes** › 2 pages
- **64 athletes** › 3 pages

Each match shows blue/red athlete names, scores split (`2+5+0+4+0=11`), and the winner. Consolation matches (3rd/4th, optional 5/6 + 7/8) live on the finals page below the main tree. Final-place medals (`1st`, `2nd`, `3rd`, `4th`, etc.) annotate the finals matches in gold / silver / bronze / gray.

Final Place List (dual mogul)

Ranked list of dual mogul athletes with their representation (club) and FFSP points. Status indicators for DSQ / DNS / SCR / DNF athletes.

Calculation Report

A detailed per-run computation breakdown. Shows the formula used (e.g., `(judge scores: 4.5 + 5.0 + 4.5) × DD 0.49 = 6.61`), so the TD can verify the engine matches the rule book on every run. Especially useful for aerials.

Calling the PDFs

Each PDF has its own endpoint under `/api/pdf/...`:

- `event-results`
- `check-sheet-bib`
- `check-sheet-runorder`
- `start-list`
- `td-report`
- `dual-bracket`
- `dual-results`
- `calc-report`

All return `application/pdf` with `Content-Disposition: attachment`. The browser downloads them automatically.

Truncation rule

All numeric values in PDFs are **truncated** (floored) to 2 decimals per FIS rule. DD values are preserved at full precision. Pace time is also truncated. This matches the engine's runtime behavior — PDFs and live scores show the same numbers.

Layout

Letter-size, portrait by default. Most reports use a 10pt body with a 14pt title block. Spaced grids and clean column headers. The compact dual bracket PDF uses a tighter 7pt body to fit the tree.

CSV / Excel exports

For spreadsheet workflows — feeding standings to a webmaster, manually computing FFSP for non-dual events, archiving — every event can be exported as CSV or Excel.

Where to access

The **Reports** tab on the event detail page has CSV / XLSX / ZIP buttons. Each triggers a server-side export.

CSV format

Tab-separated columns, with header row:

```
Place, Bib, Last, First, Club, Div, Birth Year, USSA#, Turns, Air, Time, Speed, Total
```

For multi-phase events, additional columns per phase:

```
..., Run 1 Turns, Run 1 Air, Run 1 Speed, Run 1 Total, Run 2 Turns, ...
```

For aerials v2 events:

```
..., Jump 1 Air, Jump 1 Form, Jump 1 Landing, Jump 1 Total,
     Jump 2 Air, Jump 2 Form, Jump 2 Landing, Jump 2 Total,
     Air no DD, Form Total, Landing Total, Event Total
```

Per-judge columns optionally included (per CLAUDE.md v1.7+):

```
..., TL1 Carving, TL1 Absorption, TL1 Upper, TL1 Deduction,
     TL2 ..., TL3 ..., A1 J1, A1 J2, A2 J1, A2 J2
```

Excel (XLSX)

Same data as CSV, but formatted as a proper `.xlsx` workbook with column widths, header styling, and number formatting (2-decimal display).

The XLSX export uses `exceljs` server-side. Numeric values are stored as numbers (not strings), so spreadsheet formulas work directly.

ZIP export

For multi-phase events, the ZIP export bundles:

- One CSV per phase
- One CSV with combined results
- A `README.txt` explaining each file

Useful when you need to deliver multiple views to a webmaster who'll merge them into a season aggregate.

Accented names in Excel

CSV exports include a UTF-8 byte-order mark, so Excel on Windows renders accented athlete names correctly when you double-click the file — no import wizard needed.

Nation column

As of v1.19.01, the **Nation column is removed** from CSV / Excel / HTML print exports. Earlier versions emitted an empty `Nation` column because the v1.16.12 Athletes UI removed Nation. The underlying `athletes.nation` DB column is preserved for USSS transmit XML and meet import/export round-trip — just no longer in exports.

DNS / DNF / DSQ rows

Athletes with a status (DNS / DNF / DSQ) appear in the export with `DNS` / `DNF` / `DSQ` in the Total column and blanks in the component columns. NT runs show `NT` in the Time column.

Truncation

All numeric values are floored to 2 decimals (FIS rule), matching the engine and the PDFs. DDs are preserved at full precision when included.

When to use which

- **CSV** — quick view, importing into web tools, scripting.
- **XLSX** — handoff to non-technical stakeholders, formatted reading.
- **ZIP** — multi-phase events, full archive bundle.

Direct API

```
GET /api/events/:eventId/export.csv
GET /api/events/:eventId/export.xlsx
GET /api/events/:eventId/export.zip
```

All return appropriate `Content-Type` headers and `Content-Disposition: attachment` with the right filename.

USSS transmit XML

For USSS-sanctioned events, the official results must be submitted to USSS in their **TransmitFreestyleResultsXML** format. StickIt generates one XML file per event and bundles the whole meet into a single zip.

When to transmit

Per the USSS rule book, sanctioned events must transmit results within a defined window (typically 72 hours of event completion). The Race Administrator / TD is responsible for the final transmission.

How to generate

1. Open the meet detail page.
2. Click **More** › **USSS Transmit**.
3. The popup lists every event in the meet with a readiness check ([yes] or [no]) and shows the derived category and USSS code for each.
4. Click **Generate Transmit File**. The browser downloads `SF<code>.ZIP` containing one XML per event (`SF<code>_<gender>_MO.xml`, `_DM.xml`, or `_AE.xml`).
5. **Email the downloaded file to results@ussa.org.** StickIt does not upload anything itself.

Everything is derived automatically — no category dropdown, no typed event IDs:

Value	Source
USSS code (`NAT_code`) per event	The **USSS Code** field on each event's page
Category (DIV / EQS / ROC)	The event's division (Comp Series › DIV, RQS-EQS › EQS, Devo › ROC)
Technical Delegate (DIV/DIC only)	The **Technical Delegate** in the meet's Officials list (name + USSS #)

Requirements — the transmit blocks until everything is ready

If **any** event in the meet has a problem, nothing is generated and the popup lists every issue:

- Every event must have status **Complete**.
- Every event must have its **USSS code** set (on the event page).
- Every event's division must map to a USSS category.
- Mogul / dual events need turns judges assigned; aerials events need judges assigned.
- When any event's category is DIV/DIC, a **Technical Delegate** with a USSS number must be assigned in the meet's Officials section.

Athletes missing a USSS member number are a **warning**, not a block — the popup lists them by name and bib and lets you proceed anyway.

What's in each XML

- **Race header** — season, USSS code, category, discipline (MO / DM / AE), race date, event name, place, and the TD block for DIV/DIC.
- **Results** — every athlete in rank order with USSS member number, name, birth year, nation (defaults to `USA`), bib, run scores, and total.
- **Judges panel** — scoring judges and Head Judge with USSS official numbers, plus meet officials (Chief of Competition, Chief of Score, etc.).
- **Not-classified block** — DNS / DNF / DSQ athletes (see below).

Athletes with no valid result (DNS / DNF / DSQ)

Athletes whose **every** run carried a status are not silently dropped — they appear in the XML's not-classified block (`<FS_notclassified>`) with their bib, competitor info, and a single resolved status (precedence: DSQ, then DNF, then DNS across the athlete's runs). Entries are ordered DNF, then DNS, then DSQ. A legacy `RNS` status is emitted as `DNF` — the stored data is untouched; only the XML output maps it. For dual mogul, an athlete who never won a match and was eliminated by status appears not-classified; an athlete who won at least one match before losing by status keeps their bracket placement in the classified list. All emitted status codes (`DNS`, `DNF`, `DSQ`) are FIS-standard.

Tie-break in the XML

Ranks in the XML use the FIS-compliant tie-break (see Tie-break rules), so the order matches the PDF Event Results Summary.

Nation field

Defaults to `USA` when `athletes.nation` is blank. This is preserved in the DB column even though the UI has dropped the Nation column from athletes editing. To set a non-USA nation for a foreign athlete, edit the athlete via direct API or use the Admin Athletes management.

Transmit failures

If USSS rejects the XML:

1. Open it in a text editor and look for obvious errors (missing USSS#, malformed date, wrong USSS code).
2. Fix in StickIt and re-generate.
3. Re-email the new file to results@ussa.org.

Verification

After generating, unzip and eyeball the XML files in a text editor:

- Is there one file per event, named with the right USSS code?
- Are all athletes present?
- Is the rank order correct?

- Are DNS / DNF / DSQ statuses encoded?
- Is the panel of judges complete?

Cross-check against the Event Results Summary PDF — both come from the same `rankResults` call, so they should match exactly.

Judges (Tablets)

T&L Judge tablet

The T&L (Turns & Line) judge tablet is opened by going to the URL on the event's **Links** tab (e.g., `/judge/<event-short>/<judge-short>`). One judge per tablet; the URL is tied to a specific judge role (`TL1`, `TL2`, or `TL3`).

Layout

- **AthleteBar** (top) — bib, name, run number, jump codes for the current athlete.
- **Score panel** (center) — components or single raw score, depending on event setup.
- **Reference sidebar** (right, 260px wide) — colored badges for "Excellent / Good / Adequate / Poor" with score ranges, so the judge can sanity-check their pick.
- **Run history** (bottom) — your recent submissions for this event, for self-reference.

Component scoring (Comp Series default)

Three columns of score buttons:

1. **Carving** (0–10) — 0.1 increments, color-zoned. Quick reference: 8.1–10 Excellent, 6.1–8 Good, 4.1–6 Adequate, 0.1–4 Poor.
2. **Absorption / Extension** (0–5) — same color zones, 0–5 scale.
3. **Upper Body** (0–5) — same.

Then a **Deduction** pad with five preset values: `0.1`, `0.5`, `1.0`, `1.6`, `6.0`. Plus a **Manual entry** button for custom deductions.

Computed total = `Carving + Absorption + Upper Body - Deduction`, clamped 0.1–20. Displayed live as you tap.

Non-component scoring (RQS-EQS)

Two columns:

1. **Raw Score** (0–20) — 4×11 grid of 0.5-increment buttons.
2. **Deduction** — same pad as component mode.

Total = `Raw - Deduction`, clamped 0.1–20.

Fine tune

Below each score column is a fine-tune row: `-0.1 / current value / +0.1`. Tap to nudge the picked score by a tenth in either direction.

Submitting

Once you've picked Carving / Absorption / Upper Body / Deduction (or Raw / Deduction in non-component mode), tap **Submit Score** at the bottom. The score is sent to the server. The button greens out as **Score Submitted** until the HJ approves or rejects.

If rejected

If the HJ rejects your submission, the tablet flips back to the score-entry state with your previous values cleared. Enter new values and re-submit.

If approved

Once approved by the HJ, your tablet shows **Score Approved** for a few seconds, then advances to the next athlete.

Self-review

The bottom of the tablet shows your last 5 submissions for the event. Useful for "wait, what did I just score?" moments.

Pin lock

If a PIN was set when the judge was added (Adding & assigning judges), the tablet asks for the PIN on first open. Cached locally for the day.

What you can't do

- Reject your own submission — only the HJ can reject.
- See other judges' scores — only the HJ tablet sees the full panel.
- Change a previously-finalized score — only the chief of score can, via Editing a finalized score.

High-contrast mode

Tap the HC Mode button (top-right) to switch to black/white/amber tokens for outdoor / bright-light visibility. See High-contrast (HC) mode.

Air Judge tablet

The Air judge tablet handles jump codes and per-jump air scores. Open via the URL on the event's **Links** tab (`/judge/<event-short>/<judge-short>` — same path as T&L; the role on the judge record determines the UI shown).

Layout (single-screen, two-column)

- **AthleteBar** (top) — bib, name, run number; small multi-line jump chips on the right.
- **Left column** — Jump 1 code grid + Jump 1 score grid.
- **Right column** — Jump 2 code grid + Jump 2 score grid.
- **Reference sidebar** (260px) — score range reference.

Both jumps render side-by-side so the judge can fill code/score in any order, then submit both together. Devo events (1 jump) render only the left column full-width.

Jump code grid

A quick-pick grid of frequently-used codes (three rows for Comp Series, 7×2 for Devo/RQS) plus an "All codes" dropdown for the long tail.

- **Comp Series defaults (three rows of six, v2.3.00):** `N, S, T, K, TS, 3` · `bT, bp, bL, bG, bg, bF` · `7op, 7oG, 7og` + No Jump. `bg` and `7og` are the basic-grab codes (lowercase g).
- **Devo / RQS defaults:** `S, T, D, X, K, TS, TT, TD, TTS, 3, 3p`
- **No-Jump button** — for jumps where the athlete didn't attempt one (rare; usually a fall after takeoff).

Dropdown lets you pick any code from the full chart. Filters work — type `b` to show all back-family codes.

Score grid

Mixed-step grid for air scores: `0.0, 1.0, 2.0, 3.0, 4.0, 5.0, 6.0, 6.5, 7.0, 7.5, 8.0, 8.5, 9.0, 9.5, 10.0`. The 1.0-increment chunk is for low scores; the 0.5-increment chunk for higher scores where precision matters more.

Selection anchor: when fine-tuning drifts the value off-grid (e.g., 7.3), the nearest anchor on the grid is highlighted so the UI stays comprehensible.

Fine tune

Same `-0.1 / value / +0.1` row as the T&L tablet. Drift one tenth at a time.

Submitting

Tap **Submit Score** at the bottom. The submit handler first sends the codes (`PUT /runs/:id` with `jump1_code`/`jump2_code`), then posts the scores. The first Air judge to submit sets the run's official codes.

Jump code mismatch (v2.3.00)

If your codes differ from the other Air judge's (compared **exactly**, including case — `bG` and `bg` are different jumps with different DDs), your scores are still recorded. The run keeps the first judge's codes for the moment, and the **Score Submitted** screen on **both** Air judge tablets shows a prominent red warning:

> WARNING — JUMP CODES DO NOT MATCH. Please see the Head Judge to reconcile.

Nothing is lost and nothing needs re-entering yet. The Head Judge's tablet shows a **Jump Code Mismatch** box listing each Air judge's codes, and resolves it one of two ways:

HJ action	What you see
Accept These Codes (one judge's pair)	The warning is replaced by **Air Codes Reconciled by Head Judge** with the accepted codes. Both judges' scores stand and are scored against the accepted codes' DDs.
Reject Both Codes	Both Air judges' codes and scores are cleared; your tablet returns to code entry (the usual rejected-score flow). Agree on the codes and resubmit.

The run cannot be finalized while a mismatch stands, so there is no risk of the wrong DD being published.

Per-judge codes

Each Air judge picks codes from scratch — no pre-select. If both Air judges agree, the codes match server-side and the run proceeds. If they disagree, the mismatch goes to the Head Judge as described above.

DD reference

Each picked code shows its DD next to the chip in the AthleteBar (`Jump 1: TS · 0.49`). For unfamiliar codes, refer to Jump codes & DDs on a separate device, or use the Officials sidebar's Jump DDs panel.

High-contrast mode

HC Mode button top-right. See High-contrast (HC) mode.

Aerials Judge tablet

The Aerials Judge tablet (v2 model, used by every aerials event from v1.18.00 onward) lets one numbered judge enter **Air, Form, and Landing** for **every jump** of every athlete. Each judge has their own tablet URL and their own panel slot.

Opening

The URL pattern is `/aerials-judge/<event-short>/<judge-short>`, taken from the event's **Links** tab. Each `AeJudge1..N` row has its own short code.

> The legacy URL pattern `/aerials-judge/<event-short>` (no judge short) still loads — that's the legacy aerials judge view for pre-v1.18.00 events. New events all use the per-judge URL.

Layout

- **AthleteBar** (top) — bib, name.
- **Jump 1 panel** (left half) — three components stacked:
- **Air** (0.0–2.0)
- **Form** (0.0–5.0)
- **Landing** (0.0–3.0)
- **Jump 2 panel** (right half) — same components.
- **Jump code chips** (top of each panel) — show the code the operator entered for that jump. Read-only on the aerials judge tablet (the code comes from the chief of score, not the judges).

Entering scores

For each component:

1. Tap a value on the quick-pick grid (0.1 increments for Air, 0.5 for Form, 0.2 for Landing — varies by version).
2. Use fine-tune `+/-` for off-grid precision.

You can enter Jump 1 and Jump 2 in either order. The submit button is enabled once every cell has a value.

Submitting

Two-step:

- **Submit Jump 1** — sends Jump 1's Air / Form / Landing.
- **Submit Jump 2** — sends Jump 2's.

OR a single **Submit Both** button (when both jumps are filled).

After submission, your three values per jump are locked. If the HJ rejects, the tablet returns to entry mode with values cleared.

Reduction rule (server-side)

After every judge submits, the server applies the reduction:

- **5+ judge FIS panels** — drop high + low per component independently, sum the kept three (or middle four/five for 6/7-judge).
- **2–4 judge USA panels** — operator-selected method (`sum_all`, `drop_high`, `drop_low`, `average`).

The judge tablet shows your raw input, not the reduced total — that's the HJ's view.

HJ may score

For USA Regional events with HJ-may-score enabled, the HJ tablet shows their own row in the panel grid alongside the AeJudge1..N rows. The HJ enters Air/Form/Landing per jump in addition to approving the whole athlete.

Why per-judge-per-jump

Pre-v1.18.00 aerials used component-specific judge roles (`AirJudge1` only entered Air, etc.). That was inconsistent with FIS JH 6004 / USSS 4110, which states every scoring judge evaluates all three components. The v2 redesign brings the model into compliance.

Reference

A small reference panel shows score range hints for each component:

- **Air** — height + distance off the kicker
- **Form** — body position, control, takeoff
- **Landing** — body angle, posture, impact absorption

High-contrast mode

HC Mode button top-right. See High-contrast (HC) mode.

Dual Mogul Judge tablet

The Dual Mogul Judge tablet handles one judge role in the 5-point split scoring format used for dual mogul brackets. There are five dual judge roles: `DualTurns1`, `DualTurns2`, `DualAir`, `DualTime`, `DualOverall`. Each judge has their own tablet URL.

5-point split format

In dual mogul, two athletes ski simultaneously. For each match, each judge awards five points total, split between the two athletes:

- `5–0` — total domination by one side
- `4–1` — clear win
- `3–2` — close win
- `2–3`, `1–4`, `0–5` — same on the other side

The five points are NOT cumulative — each judge picks one split per match. Five judges × one split each = a maximum of 25 points total per match (5 + 4 + 3 + 4 + 5 etc.).

Layout

- **AthleteBar with two halves** — blue side (athlete 1) on the left, red side (athlete 2) on the right.
- **Score buttons** — six large tap targets, one per split: `5–0`, `4–1`, `3–2`, `2–3`, `1–4`, `0–5`.
- **Submit Score button** at the bottom.

Entering

1. Watch the match.
2. Decide who won and by how much.
3. Tap the corresponding split (e.g., `4–1` for a clear blue win).
4. Tap **Submit**.

The tablet locks; the score is sent to the server. When all 5 judges have submitted, the HJ tablet shows the full match and asks for approval.

Match navigation

The tablet shows the **current active match only**. Use the Scoring tab's bracket view to see upcoming matches. Once the current match is approved, the tablet auto-advances to the next match.

Judge role differences

In dual mogul, the five judge roles are nominally:

- **DualTurns1** / **DualTurns2** — two judges focused on turns quality
- **DualAir** — air quality judge

- **DualTime** — time-down-the-hill judge (eyeballs, no clock)
- **DualOverall** — overall impression

All five use the same 5-point split UI — the role labels are nominal, for FIS reporting structure. In practice each judge weighs their role's component but submits one combined split. Three roles have extra controls:

- **DualAir (Judge 3)** — an **Air Tied** button (submits 0 / 0 with the air-tied flag when the air comparison is tied or neither competitor jumps; the air votes are withheld) and the **NJ (Past Chop)** panel described below. **Both are gated by the meet's Advanced settings** and are hidden by default — Air Tied appears only when "Air score tie allowed" is on, and the NJ panel only when the "Landing Past the Lower Chop (NJ) rule" is on.
- **DualTime (Judge 4)** — the **Time Tied** button. A time-tied entry records 0 / 0 but is credited **3 / 3** in every display and total (v1.29.00, FIS JH 6304.3.2 five-judge value).
- **DualOverall (Judge 5)** — the split grid automatically switches between **5, 4, or 3 points** depending on the tie state: 5 with no ties, 4 when speed OR air is tied, 3 when both are. If a tie declaration or NJ change lands after the Overall Judge already submitted, their entry is cleared and the tablet prompts a rescore on the new scale.

NJ (Past Chop) — Air Judge only

Per FS-18, the bottom air must land within the 20 m Landing Zone. **The rule is off by default** — the panel appears only when the meet's Advanced settings enable the NJ rule. When enabled, the Air Judge's tablet has an **NJ (Past Chop)** panel below the split grid with independent **Blue NJ** / **Red NJ** toggles:

1. Tap the violator's toggle. A confirmation appears — "Mark [Name] as NJ, landed past chop? This sets their speed points to zero." (If the other side is already flagged, the confirm notes the result is a speed tie at 3 / 3.)
2. Confirm. The call posts to the server immediately — it does not wait for, and is independent of, your air split.
3. Clearing a toggle also asks for confirmation. Toggles stay editable until the Head Judge approves the match.

A single NJ overrides the Time Judge's entry to **0 / 5** for the violator; both NJ makes speed a true tie at **3 / 3**. The Time Judge still enters their real split as evidence — a persistent amber banner on the Time Judge's tablet shows the active override. The Head Judge can review, set, or clear the call before approving.

Manual Score Entry lockout

When the chief of score has opened the **Manual Score Entry** modal for the current match, the judge tablets show a full-screen amber overlay:

> Manual Score Entry for This Round

Judges cannot submit during this lockout. When the operator either submits the manual score or cancels, the lockout clears and tablets resume normal operation.

Event complete

After every bracket match has been finalized (and the HJ has clicked **Approve & Finalize Event**), the judge tablets show a full-screen "Event Completed — Thank You for Your Work" message.

High-contrast mode

HC Mode button top-right. See High-contrast (HC) mode.

Timekeeper tablet

The Timekeeper tablet enters run time after each athlete finishes. URL pattern: `/timekeeper/<event-short>`. One tablet per event (not per judge — only one timekeeper).

Layout

- **AthleteBar** (top) — bib, name, run number for the currently-scoring athlete.
- **Finish time display** (88px tall, Bebas Neue) — shows the entered time as you type.
- **Custom numpad** — 3×4 calculator-order grid: `7 8 9 4 5 6 1 2 3 · 0 ·` with **Backspace** inline to the right of the time display (80px wide, red-tint border).
- **Action buttons** below the numpad:
- **NT** (No Time) — red, left.
- **Submit Time** — green, flex 2.5, right. [yes] checkmark prefix.
- **Manual Time Calculation (Top / Bottom Timer)** — amber/orange, below.
- **Right sidebar**:
- **Next Up** card with athlete + Start Run button (when no athlete is in scoring state).
- **Previous Times** — list of recent times, color-coded (green for under pace, amber for slower).
- **Pace Time** box — the meet's pace time, for sanity check.

Entering time

1. Tap digits on the numpad to build the time as a decimal (e.g., `27.34`).
2. Tap backspace if you mistype.
3. Tap **Submit Time** to send.

The time is associated with the currently-scoring athlete. The server records it on the run row; the HJ tablet shows time complete.

NT (No Time)

Tap the red **NT** button if no valid time could be recorded but the athlete did finish. A confirmation modal appears:

> Mark [Athlete Name] as No Time?

Confirm to submit. The run gets `run_time = -1` (sentinel); the speed score is set to 0. The Time column on every surface shows "NT". See Run statuses.

Manual Time Calculation

For events using a top timer + bottom timer setup (each measuring a portion of the course), tap **Manual Time Calculation (Top / Bottom Timer)** to open a modal:

- Enter the **Top time** (start gate to mid-course)
- Enter the **Bottom time** (mid-course to finish)

- The modal computes total time = top + bottom
- Confirm to submit

Useful when no single full-course timer is available.

Start Run

When no athlete is currently scoring, the **Next Up** card on the right side shows the next athlete with a blue **Start Run** button + yellow **DNS** button. Tap Start Run to begin scoring that athlete (`runs.status='scoring'` is set, all tablets see the new athlete).

If `score_entry_mode='paper'`, the Start Run button stays visible even mid-scoring (so the timekeeper can start subsequent runs as the operator).

The Start Run and DNS buttons appear only when the meet's Advanced settings allow the Timekeeper to start runs (on by default). Pressing Start Run first re-checks the server's next-up athlete — a card that went stale while the iPad was asleep can never start the wrong athlete — and the server refuses duplicate starts and (in tablet mode) starts while another run is still scoring.

Age group transition banner

When the next athlete is in a different age group from the just-finished one, an amber banner appears at the top:

> U11 group complete — U13 up next

Auto-dismisses after 15 seconds or on tap. Devo events only.

Devo events — no time

Devo events have `has_speed=0`, so the timekeeper tablet is **not used**. The event has no Timekeeper role.

Dual mogul — no timekeeper

Dual mogul has its own dual judge role (`DualTime`) that uses the dual judge tablet, not the timekeeper tablet. The standalone timekeeper tablet is for standard mogul / aerials only.

Event complete

After all phases are finalized, the tablet shows the full-screen "Event Completed — Thank You for Your Work" message.

High-contrast mode

HC Mode button top-right. See [High-contrast (HC) mode](#).

Head Judge tablet

The Head Judge tablet is the oversight surface during live scoring. The HJ doesn't enter scores — they review what the other judges submitted, approve or reject individual submissions, mark run statuses (DNS / DNF / DSQ), and finalize phases and events.

URL

`/headjudge/<meet-short>/<event-short>` — taken from the event's **Links** tab. Unlike judge tablets, the HJ URL doesn't include a per-judge short code (there's only one HJ per event).

Layout

- **AthleteBar** (top) — bib, name, run number, jump codes.
- **Status squares** (3-column status bar) — `T&L Judges 3/3`, `Air Judges 2/2`, `Time [yes]`. Green when complete, gray when pending.
- **Per-judge review rows** — each row shows the judge's role, their submitted score, and a small **Reject** button.
- **Calculated score panel** (right) — Turns, Air, Speed/Time Points, Total. Computed live as judges submit.
- **Run Status grid** (bottom) — DNS / DNF / DSQ buttons.
- **Approve & Submit button** (large green, bottom) — appears only when every required judge has submitted.

Per-judge review rows

For each judge, the row shows:

- **T&L judges** — component breakdown (`Crv 5.5 / UB 1.0 / A&E 4.5 / Ded 1.5`).
- **Air judges** — per-jump scores with jump codes (`J1: TS 0.49 › 3.4 / J2: bp 0.62 › 4.5`).
- **Timekeeper** — entered time, computed time points.

A red **Reject** button at the right of each row opens a confirm dialog before clearing that judge's submission. Above the Air rows, **Reject Codes** clears the run's jump codes together with both Air judges' scores.

Jump Code Mismatch box

When the two Air judges submit different jump codes (compared exactly — `bG` and `bg` are different jumps), both judges' scores are still recorded and a red **JUMP CODE MISMATCH** box appears at the top of the Air Judges card, one line per Air judge with their codes (`bT / bG`, `bT / bp`) and a green **Accept These Codes** button on each line, plus **Reject Both Codes** on the bottom row.

- **Accept These Codes** makes that judge's codes the run's codes for both Air judges (DDs resolve from the DD table exactly as if that judge had entered them first). Both Air judge tablets replace their warning with **Air Codes Reconciled by Head Judge**. The Running Score updates to the accepted DDs.
- **Reject Both Codes** is the same action as **Reject Codes**: both Air judges re-enter codes and scores.

Finalize and Publish Score stays disabled while the mismatch stands (the waiting banner reads "Jump code mismatch — reconcile below"), and the server refuses approval too. See HJ review for the full flow.

Aerials v2

For aerials v2 events, the HJ tablet shows a panel grid: rows = scoring judges, columns = (J1 Air/Form/Land, J2 Air/Form/Land). Each cell shows that judge's entered value. Computed totals appear at the bottom.

Run Status

Three big buttons: **DNS**, **DNF**, **DSQ**. Each opens a confirm dialog:

> Mark [Athlete Name] as DNS?

before submitting. See Run statuses for what each does.

Approve & Submit

Once all judges (and timekeeper) have submitted, the big green button appears. Tap it:

- Finalizes the run (`runs.status='complete'`).
- Computes ranking.
- Broadcasts `score_update` to all viewers.
- Advances to the next athlete in run order.

The score set must be complete before a run can be published. The **Finalize and Publish Score** fallback button stays disabled — with a "Waiting for scores: T&L 2/3 · Time pending" breakdown — until every required judge score (and time, when the event is timed) is in. On approval the server recomputes the total from the stored judge scores, so a rejected-then-resubmitted score always lands in the published number. Use DNS/DNF/DSQ for an athlete who did not complete the run.

Starting runs from this tablet

The **Start Run** button (with its DNS companion) on the next-up card appears only when the meet's Advanced settings allow the Head Judge to start runs (on by default). When pressed, the tablet re-checks the server's next-up athlete first — so a card that went stale while the iPad was asleep can never start the wrong athlete — and the server refuses duplicate or concurrent starts outright.

Send Back to Scoring

Amber button. Opens the whole run back to specific-judge editing — useful when multiple things need to be corrected.

Phase summary

After every athlete in the current phase has been approved, the HJ tablet shows a **Phase Summary** card with:

- Combined per-athlete list (Run 1 / Run 2 columns for Best of 2).
- **Approve Phase** button — locks the phase, advances to next phase.
- **Send Back to Scoring** button — re-opens any run for editing.

Final Review

After every phase is finalized, the **Final Review** screen appears: a combined table of all athletes with their per-phase scores, plus:

- **Approve & Finalize Event** (green) — sets `events.status='complete'`. The full-screen "Event Complete — Thank You for Your Work" appears.
- **Send Back to Scoring** (amber).

Dual mogul HJ

Dual mogul HJ tablet manages match-by-match approval. After all bracket matches are complete, the bracket-review panel appears with the full bracket tree and the same two buttons.

Set Match Status (Blue/Red DNS · DNF · DSQ). Each button opens a confirmation first — "Record Blue DNF for [Name]? Red advances." When judge points already exist on the match, the confirmation escalates and shows the points state (e.g. "5 judges have scored this match (Red leads 18–7)"), with the strongest red warning when the ruling contradicts the points winner. The Head Judge's ruling has final say and stays possible even with all five judges scored; every manual ruling is audit-logged with the points state at the time. Once a match is complete and the bracket has advanced, the tablet can no longer change it — post-completion changes go through the operator's **Edit Scores** path on the Scoring tab.

For the FS-18 landing zone (chop) rule (only when the meet's Advanced settings enable it — off by default), the dual HJ view adds:

- an **NJ (Past Chop)** panel with per-athlete Set/Clear toggles (with confirmation) — the Air Judge normally makes the call, but the HJ can set it on close calls or clear a mistaken one;
- a **persistent amber banner** while a call is active, describing the speed override (e.g. "Speed override active: Blue 0 / Red 5" or "Speed tied at 3 / 3");
- **raw vs effective values** in the judge grid — the Time Judge's recorded entry is shown with the overridden value beside it (e.g. `4 / 1 › 0 / 5 (NJ)`; a time-tied entry shows `Time Tied › 3 / 3`; an air-tied entry shows `Air Tied (0 / 0 — votes withheld)`).

Approving the match certifies the NJ finding along with the scores; the call locks at approval. Clearing a finding needs no score restoration — the Time Judge's untouched entry simply governs again. Rejecting the Time Judge's entry (or an air-tied Air Judge entry) also clears the Overall Judge, since their point scale depends on the tie state.

High-contrast mode

HC Mode button top-right. See High-contrast (HC) mode.

High-contrast (HC) mode

Every tablet (Judge, Air Judge, Aerials Judge, Dual Mogul Judge, Timekeeper, Head Judge) has a **HC Mode** toggle in the top-right corner. Tapping it switches the tablet into a high-contrast palette — black background, white text, amber accents — designed for outdoor visibility in bright sunlight, on snow, or with reflective glare.

What changes

- **Background** › solid black
- **Text** › white, with thicker stroke
- **Accents** › bright amber (`#f5c518`) instead of navy/blue
- **Status squares / glow effects** › switched from green-on-dark to amber outline
- **Score buttons** › outlined white-on-black with amber selection
- **Component zones (Excellent/Good/Adequate/Poor)** › still color-coded but with HC-safe contrast ratios

The HC palette is **functionally identical** to normal mode — all the same buttons, same layout, same workflow. Only the visual tokens change.

Toggle button

A small button labeled `‹ Normal Mode` (when HC is on) or `HC MODE ON` (badge state) — appears in the same place on every tablet. Tapping it flips the mode and persists the choice in `localStorage.stickit-high-contrast`.

Persistence

The choice is local to that tablet. Each judge's tablet has its own HC preference. If you toggle on one tablet, it doesn't affect any other tablet.

When to use

- **Bright sunny day on snow** — sunlight + glare make the normal navy palette hard to read.
- **Old or low-brightness iPad** — older display backlights wash out in daylight.
- **Aging eyes** — the higher contrast is easier on the eyes for long scoring sessions.
- **High-contrast accessibility need** — for judges with visual impairments.

When NOT to use

- **Indoor / dark conditions** — the HC palette is harsh in dim light. Normal mode is gentler.
- **For broadcast / video review** — HC mode looks unfamiliar on camera; reserve for actual judging.

Status squares

Status squares are slightly larger in HC mode (14×14 px vs. 12×12 in normal mode) so they stay visible at HC's increased thickness.

Reference panel pills

The Excellent / Good / Adequate / Poor reference pills use HC-friendly colors (deeper saturation, white text borders) that maintain readability while still color-coding the score zones.

Devices that need HC

The standard scoring setup uses iPads at 100% brightness. Even at full brightness, a bright sunny Colorado morning can wash out the normal navy palette. Testing has shown that HC mode is reliably readable in conditions where normal mode struggles.

Not in the legacy aerials view

The pre-v1.18.00 legacy aerials judge tablet doesn't have HC mode. All other tablets do.

Underlying mechanism

HC mode is implemented as a `data-hc="1"` attribute on the tablet's root container. Every `tablet-*` CSS class has a corresponding `[data-hc="1"]` override. A legacy `.hc` class also still applies to inner Tailwind classes that haven't been migrated to the token system — belt-and-suspenders.

Admin Panel

Admin Dashboard

The Admin Dashboard is the system administrator's landing page at `/admin/dashboard`. It surfaces server health, database stats, recent activity, and the in-memory error log at a glance. Auto-refreshes every 30 seconds.

Sections

Key metrics

- Server uptime (since process start)
- App version
- WebSocket connection count (live)
- System IP + port

Run statistics

- Active runs (blue pulse) — currently being scored
- Pending runs (yellow) — waiting on HJ approval
- Complete runs (green) — finalized

Database

- Counts: meets, events, athletes, registrations, users
- DB file size
- Last backup timestamp
- Backup count

Disk space

- Progress bar showing used / total / percentage of the data partition

Error log

- Collapsible table of in-memory server errors since startup (max 100 entries, ring buffer)
- Each entry shows timestamp, tag (e.g., `BACKUP / auto`, `HTTP 500 /api/runs`), and message
- Errors are pushed by the error-capturing Express middleware and the auto-backup error callback

Recent activity

- Collapsible table of the last 20 audit log entries, with a link to the full Audit log
- Shows what was changed and when

Environment indicators

- Whether **password protection** is enabled (see Password protection & login)
- Whether a **voice service key** (`DEEPGRAM_API_KEY`) is configured — required for Voice manual entry
- Whether a fixed **JWT secret** (`STICKIT_JWT_SECRET`) is set via environment (optional; StickIt persists its own otherwise)

Server endpoint

`GET /api/admin/dashboard` returns all the data in a single call. The page polls every 30s; manual refresh is also available via the **Refresh** button.

What's not yet here

- Per-event drilldown (planned)
- Detailed query traces (out of scope; consider the host dashboard — Render/Railway — or Grafana)
- Email alerts on errors (out of scope)

Sidebar entry

The Admin sidebar's first entry is **Dashboard** (`/admin/dashboard`). The route `/admin/system` (legacy `System Info`) now redirects to the dashboard.

When to check

- **Beginning of meet day** — verify server uptime, free disk, no pending errors, backup recent.
- **Mid-day during a meet** — quick sanity check after a connectivity event ("did the WebSocket count drop?").
- **End of meet** — review error log for anything that warrants follow-up.
- **Before a big release** — full review of activity, errors, DB stats.

Setup tracking

The dashboard captures `app.startedAt`, `app.errorLog`, and `app.wss` references registered in `server/index.js` at boot. If you're running a custom server fork, ensure those are wired or the dashboard fields will be blank.

Password protection & login

StickIt ships with password protection **off** so a brand-new install works immediately. When you're ready to put the server on an open network (or the internet), turn protection on from **Admin › Security**. With protection on, every Officials and Admin page requires a login; the public and judging surfaces keep working without one.

What stays public — always

These are never behind the login, by design:

- **Judge / Head Judge / Timekeeper / Aerials tablets** and all their scoring endpoints — secured only by their unguessable short-code URLs. Enabling protection mid-meet can never lock judges out.
- **Public pages** — Home, Live Scores, Scoreboard, the broadcast Overlay, and this Help guide.
- The PDF downloads reachable from the public Scoreboard (detailed results, dual bracket, dual place list).

What the login protects

- The entire **Officials** section — meets, events, registration, scoring, exports, training days, USSS transmit.
- The entire **Admin** panel (System Admin role required).

Roles

Role	Can do
Judge	Log in; Officials dashboard restricted to the Links area
Official	Full Officials section — run meets end-to-end
System Admin	Everything, including the Admin panel

Roles are managed on the User management page.

Enabling protection

1. Go to **Admin › Users** and set a password on at least one **System Admin** account. The Security page will refuse to enable protection until an admin password exists — otherwise you'd lock yourself out.
2. Go to **Admin › Security** and click **ENABLE PASSWORD PROTECTION**.
3. The Home page divider switches to **LOGIN REQUIRED**, and the Officials / Admin buttons route through the login page.

Disabling works the same way — the **DISABLE PASSWORD PROTECTION** button on the same page.

Sessions & login behavior

- Login sessions last **12 hours**, and survive server restarts and redeploys.
- Repeated failed logins are throttled: 10 failures for the same username or address within 15 minutes blocks further attempts temporarily.

Tips

- Check the Users table's **Password** column (User management) to see who has a password set before enabling protection.
- The Admin Dashboard shows whether a fixed JWT secret is configured via environment variable; without one, StickIt generates and persists its own automatically.
- If you ever lock yourself out, protection can be disabled at the server (environment `STICKIT_AUTH=off`) while you fix accounts.

User management

The User Management page at `/admin/users` is full CRUD for StickIt login accounts. Accounts matter once [password protection]() is enabled — with protection off, the app doesn't ask anyone to log in, but you can stage accounts and passwords ahead of time.

Roles

Three roles, lowest to highest:

Role	Access
Judge	Login only; the Officials dashboard is restricted to the Links area
Official	Full Officials section — meets, events, registration, scoring, reports
System Admin	Everything, including this Admin panel

A higher role can do everything a lower role can. (Accounts created in older versions with the legacy `event_admin` role are automatically migrated to System Admin.)

The users table

Each row shows username, display name, role, a **Password** column (Set / Not set), and active status. The Password column is the quick pre-flight check before enabling protection — anyone marked **Not set** cannot log in.

Adding a user

1. Click **+ New User**.
2. Fill in username, display name, role, and a password.
3. Click **Create**.

Editing

Click **Edit** on any row to change display name, role, active status, or to set / reset the password. Usernames cannot be changed (they're the de facto key).

Deactivating

Click **Deactivate** (a confirmation is required). Sets the account inactive — the user can no longer log in, but the row remains so audit history stays intact. Two guard rails protect you here:

- You cannot deactivate **your own** account.
- You cannot deactivate the **last active System Admin** — there must always be at least one admin who can get back in.

Audit log

User creates, edits, and deactivations are logged. View on the Audit log page.

Bulk user import

There's no bulk import. For an organization onboarding many users, add them one at a time via the form.

Event Management (Admin)

The Admin › Events page at `/admin/events` is where every event in the system can be **locked**, **hidden from Live Scores**, or **re-opened** after finalization. It's also the canonical view of all events across all meets — a useful audit lens.

Meet-grouped layout

Events are grouped by meet. Each meet appears as a collapsible section:

- **Header row** — meet name + date, with status badges when collapsed ("All locked" / "Partially locked", "Hidden from Live Scores" / "Partially hidden").
- **Lock All / Unlock All** and **Hide All / Show All** bulk buttons (each asks for confirmation).
- A **search box** above the list filters by meet or event name.

Expand a meet to see its event rows: event name, discipline, status, a Lock indicator, a Live Scores indicator (Visible / Hidden), and the action buttons.

Locking an event

Click **Lock** on a row. Immediate effects:

- The event becomes read-only — every mutation endpoint is refused.
- The event is hidden from the Officials Dashboard (and the whole meet disappears from the Dashboard if every event under it is locked).
- Public scoreboards and the broadcast overlay continue to display the final state — locking is invisible to the public.

Locking is **reversible and non-destructive**; the data stays intact.

When to lock: end-of-meet after PDFs and USSS XML are out; pre-archive; or mid-meet to protect a finalized event while later events are still running. **When to unlock:** a TD-approved correction, a re-opened protest, or fixing a typo.

Hiding an event from Live Scores

Click **Hide** on a row (or **Hide All** on a meet) to remove the event from the public **Live Scores** page and from third-party apps using the public event list. This is the tool for keeping **test events** out of public view.

- Hiding is **completely independent of locking** — an event can be locked, hidden, both, or neither.
- Hiding affects *listings only*. Anyone with the direct scoreboard short-code link can still view the event, the broadcast Overlay keeps working, and judge tablets are unaffected — so you can fully test a hidden event.
- If **every** event in a meet is hidden, the meet itself disappears from the Live Scores page.
- New events are always visible by default. Click **Show** (or **Show All**) to restore visibility.

Re-opening a finalized event

Events with status **complete** show a **Re-open** button. Click it (and confirm) to set the event back to *in progress* so scores can be corrected — for example after an upheld protest. For dual mogul events this also clears the bracket's Head-Judge approval so the review flow runs again. The event must be finalized again afterward. Every re-open is recorded in the Audit log.

Bulk actions

Per meet, **Lock All / Unlock All** and **Hide All / Show All** toggle every event under the meet in one click — useful at end-of-meet, or to drop a whole test meet off Live Scores.

Compare with the Officials sidebar

The Officials Dashboard automatically filters out locked events — they disappear from the operator's view but stay visible here in Admin. This is intentional: Admin should be able to see and manage everything; Officials should be focused on what's actively being scored.

USSS People database (Admin viewer)

The Admin › USSS People page at `/admin/usss-people` is a viewer for the imported USSS People file (master USSS roster), with **Sync Now** and **Upload File** buttons in the status card header for refreshing the data directly.

Status card

At the top:

- **Last imported** — timestamp + source (`auto-sync` / `manual-upload`)
- **List year + identifier + filename** (e.g., `2026 Fall — people.txt`)
- **Total record count** — every record across all three types
- **Breakdown by type**:
- **Competitors (C)** — licensed competing athletes
- **Coach/Comp (CO)** — typically U15+ athletes with dual coach/competitor credentials
- **Officials (O)** — judges, TDs, course officials

Type label clarification

The USSS type code `CO` is labeled **Coach/Comp** here (not "Coach") because the USSS People File assigns CO to U15+ athletes who hold dual coach/competitor credentials — most are still primarily competitors. Pre-v1.16.20 versions used the misleading "Coach" label.

Search and filter

- **Search input** (debounced 300ms) — matches `last_name`, `first_name`, `ussa_id`, or `club_name` (LIKE prefix).
- **Type filter dropdown** — All / Competitors / Coach/Comp / Officials.

Both filters reset to page 1 on change.

Paginated table (100 per page)

Columns:

Column	Notes
USSA ID	The membership number
Last	
First	
Type	C / CO / O
Div	Division (RM, SE, etc.)
Gen	Gender
YOB	Year of birth
Club	Club name from the People file

Column	Notes
AE Pts	Aerial points
DM Pts	Dual mogul points
MO Pts	Mogul points
FIS ID	If present

Numeric columns are right-aligned. Points display to 2 decimals.

Download CSV

The **Download CSV** link streams the full table (no pagination, all records, current filter applied) as CSV. Filename pattern: `usss_people_<year>_<identifier>.csv`. Disabled when total = 0.

Endpoints

- `GET /api/admin/usss/people?q=&type=&page=&limit=` — paginated/filtered JSON (default 100 per page, max 500).
- `GET /api/admin/usss/people/download` — full CSV stream.

When to use this

- **Verify sync worked** — count of competitors should match what you expect for the current season.
- **Look up a specific athlete** — when their info on a registration looks off.
- **Audit divisional data** — filter by division to see how many athletes are registered per division.
- **Pre-meet check** — confirm the People file is current (last sync date should be recent).

Re-syncing or uploading

Click **Sync Now** (auto-fetch) or **Upload File** (manual upload of a `people.txt` you downloaded) in the status card header. The same controls also live on the Officials side at **USSS Database** (`/dashboard/usss`) — both update the same data.

Athletes database management

The Admin › Athletes page at `/admin/athletes` is for bulk-cleaning up the master Athletes database. Over time, retired athletes, manually-entered names, and one-off CSV imports accumulate; this page provides four bulk-cleanup operations.

Soft-delete model

Deletes from this page are **soft-deletes**. The athlete row is not removed from the database — only the `athletes.deleted_at` column is set. Effects:

- Soft-deleted athletes disappear from the master list, search, registration pickers, and CSV reconcile flows.
- **Past events are never affected** — existing registrations and runs still display the athlete name and any historical scores.

Auto-restore

If a soft-deleted athlete is re-added via **Add from USSS Database** or re-imported via SkiReg CSV, the existing row is restored (`UPDATE athletes SET deleted_at=NULL`) instead of creating a duplicate. This is why a CSV import after a bulk delete just brings the right athletes back automatically.

Four bulk operations

1. **Delete ALL Athletes** (red) — soft-delete every active athlete. Use after a full season to start fresh.
2. **Delete Selected Athletes** (red) — checkbox table; bulk soft-delete the chosen rows.
3. **Delete by Division** (dark) — dropdown of distinct divisions; soft-delete all in the picked division (or the "(No Division)" bucket).
4. **Delete Non-USSS Athletes** (dark) — soft-delete every athlete whose `ussa_num` is NULL/blank OR whose USSA# is not in the `usss_people` master file.

All four operations show a confirmation modal with the count and a sample of athletes before executing.

Viewing and restoring deleted athletes

Flip the **Show deleted** toggle to list soft-deleted athletes instead of active ones. Select any of them and click **Restore Selected** to bring them back to the active list — no re-import needed. Restores are audit-logged.

Page layout

- **Status card** — active total, division count, selected count.
- **Bulk action toolbar** — the four buttons.
- **Search + division filter**.
- **Checkbox table** — Last, First, USSA #, FIS ID, Club, Division, YOB, Gen, USSS column showing [yes] when found in `usss_people`.
- **Pagination** — 100 per page.

Endpoints

All require admin auth:

- `GET /api/admin/athletes?q=&division=&page=&limit=` — paginated list with `is_in_usss` flag.
- `GET /api/admin/athletes/divisions` — distinct divisions + count for "(No Division)" bucket.
- `POST /api/admin/athletes/preview-delete` — `{ mode, ids?, division? }` returns `{ count, sample }` for the confirmation modal.
- `POST /api/admin/athletes/delete` — same body, performs the soft-delete. Returns `{ deleted }` count. Audit-logged as `athletes_bulk_deleted`.

When to use each operation

- **Delete ALL Athletes** — end of season, full reset.
- **Delete Selected** — small batch of obvious duplicates or test data.
- **Delete by Division** — when a division spun off or merged and you're consolidating.
- **Delete Non-USSS Athletes** — pre-season cleanup; any athlete who isn't licensed isn't going to compete.

Don't fear the delete

Soft-delete is reversible three ways — the **Show deleted › Restore Selected** flow on this page, re-adding via the USSS picker, or re-importing via SkiReg CSV. Past events are unaffected. The only thing you lose by deleting is visibility in the master list; the data is still there if you need it back.

Hard-delete (real removal)

There's no hard-delete UI. To genuinely remove an athlete row from the database, hand-edit the SQL — not recommended in any normal workflow.

Backups

The Admin › Backups page at `/admin/backups` displays the automatic rolling SQLite backups created by `server/db/autosave.js`. Backups are read-only here — actual restore is a server-side file swap.

Auto-backup mechanism

Since v1.19.01, auto-backup is **time-based**:

- Every **5 minutes**, the auto-backup job fires.
- If at least one DB write occurred since the previous backup, a new backup file is created.
- If no writes occurred, the job skips silently.
- The system keeps the **last 10** backup files, deleting the oldest when a new one would exceed.

Backup files are timestamped `.db` files in `data/backups/` (e.g., `scoring_20260513_142318.db`).

Status card

The page displays:

- Backup count (current / max)
- Total disk used by backups
- Newest / oldest timestamps with relative-ago labels (e.g., "3 minutes ago")
- Write counter (total writes since server start)
- Backup interval ("Every 5 min if writes occurred")

Manual backup trigger

Click **Create Backup Now** to invoke `doBackup()` directly, outside the every-5-minute cycle. Useful before risky operations — e.g., before re-seeding a dual bracket, before a meet import overwrite.

Backups table

Column	Description
Filename	Monospace, full filename
Created	Timestamp + relative ago
Size	KB or MB
Download	Streams the `.db` file

Newest first. Each Download link streams the SQLite file with `Content-Disposition: attachment`. The filename regex is validated server-side (`^scoring_[\w-]+\.db$`) to prevent path traversal.

Restoring from a backup (in-app)

Each row in the backups table has a **Restore** action:

1. Click **Restore** on the backup you want.
2. Type the backup's **filename** exactly into the confirmation box — this is deliberate friction, since restore replaces the live database.
3. Before copying anything, the server takes a **pre-restore safety backup** of the current database, so even a mistaken restore is recoverable.
4. The backup is copied over `data/scoring.db`. **Restart the server** to load the restored data — the page warns you of this.

All connected tablets and scoreboards will refetch from the restored database after the restart.

Restoring manually (alternative)

The file-swap path still works if you prefer it: stop the server, replace `data/scoring.db` with the downloaded backup file, restart. On a cloud host (Render/Railway) this requires access to the persistent disk; the in-app Restore button avoids that need.

Recovery instructions on-page

The Admin › Backups page shows an amber static callout documenting the restore procedure.

Auto-backup failure handling

If an auto-backup fails (disk full, file system error), the error is captured in the `app.errorLog` in-memory ring buffer with the tag `BACKUP / auto`. It appears on the Admin Dashboard error log with the message "Auto-backup failed (covered N writes)."

Endpoints

- `GET /api/admin/backups` — returns `{ backups, stats }`.
- `POST /api/admin/backups/create` — manual backup trigger.
- `GET /api/admin/backups/:filename/download` — streams the `.db` file.
- `POST /api/admin/backups/:filename/restore` — in-app restore (takes a safety backup first).

Best practice

- **Before any risky operation** (re-seed, force-finalize, manual SQL): trigger a manual backup.
- **End of meet day**: download a backup for off-site storage.
- **Pre-deploy**: ensure a recent backup exists locally before pushing changes (pushes auto-deploy to the cloud hosts).

Manual exports separate from backups

Auto-backups are full DB snapshots — every meet, every athlete, every score. For individual-meet exports (the kind you'd share with another organization), use the per-meet Exporting a meet flow instead. Backups are server-state snapshots; exports are portable meet bundles.

Audit log

The Admin › Audit Log page at `/admin/audit` displays every system mutation captured in the `audit_log` table. Reads are not logged — only changes. Use this to investigate "who did what when" questions.

What's logged

- **Run scoring** — score submissions, edits, rejections, manual entries, status overrides.
- **Registration** — athletes added / removed / scratched / edited.
- **Run order** — random builds, age-group builds, manual reorders, lock toggles.
- **Bib assignment** — every bib update, including bulk operations.
- **Judge management** — add / edit / delete.
- **Event config** — phase changes, course spec updates, panel size changes, lock toggles.
- **Meet operations** — create / edit / import / export / delete.
- **User management** (Admin) — create / edit / deactivate.
- **Security** — password changes and enabling / disabling password protection.
- **Admin event actions** — re-opening finalized events, hiding / showing events on Live Scores.
- **USSS sync** — sync runs, upload events.
- **Athletes bulk operations** — bulk soft-deletes and restores.
- **Backups** — manual backup triggers, auto-backup failures.

Columns

Each row shows:

- **Timestamp** — UTC, displayed in local time
- **Action** — short identifier (e.g., `run_finalized`, `athletes_bulk_deleted`, `event_reopened`)
- **Entity** — the kind of thing changed (event, run, athlete, meet…)
- **ID** — the specific record's identifier
- **Detail** — JSON payload with old / new values

Filtering

Filters at the top of the page:

- **Entity** and **Action** — dropdowns populated live from the values actually present in your log, so they always match your data.
- **From / To dates** — limit the result to a date range.
- **Row limit** — how many entries to load.

Combine filters with AND logic.

Inspecting a row

Click any row to expand the **Detail** JSON. For score edits, you'll see the before/after values for every changed field. For bulk operations, you'll see the count + a sample of affected rows.

Touch / mouse support

As of v1.16.13, rows are touch-responsive on iPad. Tap to expand; tap again to collapse.

Where it moved

In v1.16.15, Audit Log was moved from `/dashboard/audit` (Officials) to `/admin/audit` (Admin). The old URL redirects to the new one.

Retention

There's no automatic retention policy. The `audit_log` table grows indefinitely. Periodically check the [Admin Dashboard] for DB size; if it grows unwieldy, an admin can hand-prune rows older than a season via SQL. A retention UI is on the future roadmap.

What's NOT logged

- Read access — viewing pages, opening modals, viewing PDFs.
- Individual sign-ins — login attempts aren't logged (repeated failures are rate-limited at the server instead).
- Backup file creations (success path) — only failures are logged.
- WebSocket connection events.
- Server boot / shutdown events.

When to consult

- **Score dispute** — "this athlete's score was 76 last night but it's 74 now — who changed it?"
- **Missing data** — "we had 18 athletes registered but now I only see 16"
- **System investigation** — "the database changed sometime around 3pm yesterday; what happened?"
- **Pre-release verification** — auditing your own changes after a setup session.

Public Surfaces

Home page

The Home page at `/` is the public landing page — the first thing anyone sees when they visit StickIt. It's intentionally minimal: branding, one primary call-to-action, and a few secondary entry points for officials and admins.

Layout

- **Translucent mountain background image** at 0.18 opacity, mix-blend-mode screen.
- **Big StickIt logo** with drop shadow.
- **"FREESTYLE SCORING" tagline** in tracking-widened Bebas Neue.
- **Primary CTA — Live Scores** — large red-gradient button. Click takes you to `/livescores`.
- **Divider** — reads **LOGIN REQUIRED** when password protection is enabled, otherwise **OFFICIALS & STAFF**.
- **Two secondary buttons** — Officials (`/dashboard`) and Admin (`/admin`).
- **Footer** — `© Rocky Mountain Freestyle` + current version (fetched from `/api/version`).

Why the Live Scores CTA is so prominent

Most visitors to a StickIt instance are spectators looking for live results. The home page funnels them straight to `/livescores` with one tap. The Officials and Admin buttons are visually de-emphasized — they're for the small group of operators.

Login behavior

The home page itself has no authentication, and the Live Scores button leads to public surfaces (also no login). When password protection is enabled, the Officials / Admin buttons route through the login page; when it's off, they go straight in.

Sun Mode toggle

A small floating sun/moon icon (top-right) lets the visitor switch between **dark mode** (default) and **Sun Mode** (high-contrast for bright outdoor viewing). The choice is saved in `localStorage.stickit.sunMode` and propagates to `/livescores` and `/scoreboard/<short>` automatically.

A small inline `<script>` in `client/index.html` runs before React mounts and paints the `<html>` background to the right color, eliminating the brief flash-of-unstyled-content when Sun Mode is on.

Version display

The footer fetches `/api/version` and shows the current version below the copyright. Updates automatically on deploy.

Branding

- **Font:** Inter Tight / Barlow Condensed / JetBrains Mono (loaded via Google Fonts CDN).

- **Logo:** `client/public/images/stickit-logo-home.png` and `client/public/logo.png`.
- **Background image:** `server/public/images/homepage-bg.jpg`.

When to bookmark

- **Operators** — bookmark `/dashboard` instead so you skip the home page entirely on opening.
- **Spectators** — bookmark `/livescores` to skip the home page.
- **Broadcasters** — bookmark the specific `/overlay/<event-short>` URL you're feeding into OBS.

The home page exists for first-time visitors and people who land on the bare domain.

Live Scores listing

The Live Scores page at `/livescores` is the directory of every meet and event in the system. It's the spectator's hub — pick a meet, drill into an event's scoreboard.

Layout

- **Sticky top header** — back-to-home + page title + division filter dropdown.
- **LIVE NOW strip** (red gradient) — currently-active events, grouped by meet. See below.
- **Meets list** — every meet, sorted latest-date-first. Each meet shows as a collapsible panel listing its events.
- **Pagination** at 10 meets per page.

LIVE NOW strip — grouped by meet (v1.19.02+)

When events are actively being scored, they appear in a red strip at the top of the page. As of v1.19.02 the strip groups by **meet** rather than emitting one card per event:

- **One red gradient card per meet** with active events.
- **Meet name** is the card header (sk-display, large).
- **One small white-tinted pill per active event** inside (showing e.g. `Men · Moguls`).
- Each pill links to that event's `/scoreboard/<short>`.

This keeps the strip compact even when three concurrent meets each have multiple active events. Single-event meets render with the same meet-card chrome for visual consistency.

Meets list

Each meet shows as a collapsible card:

- **Header row** — meet name (large), location, date.
- **Expand chevron** — collapsed by default. Click to reveal the event list.
- **Event rows** (when expanded) — discipline, gender, status (active / complete / setup), link to scoreboard.

Status indicators

- **Active** — green pulse dot. Currently being scored.
- **Complete** — blue badge. Finalized.
- **Setup** — gray. Not yet started.

Division filter

Top-right filter dropdown. Options:

- All divisions
- RM (Rocky Mountain)

- SE (Southeast)
- IM (Intermountain)
- ... etc.

Filter narrows to meets and events with athletes from the selected division.

Backend

`GET /api/meets/livescores?page=&limit=10&division=` returns a paginated list of meets with nested event arrays. Each event includes status and a link short code.

Auto-refresh

The Live Scores page silently re-fetches every 45 seconds, so the LIVE NOW strip and event statuses stay current on an unattended display. No manual refresh needed.

Sun Mode

The Sun Mode toggle inherits from the home page choice (stored in `localStorage.stickit.sunMode`). High-contrast palette for outdoor viewing.

Backlinks

Footer of each meet card: "View all events" › expands the meet. Footer of each event row: links to `/scoreboard/<short>`.

Mobile

The page is responsive — meet cards stack vertically on narrow viewports. The LIVE NOW strip wraps. Tested on iPhone portrait.

What's NOT shown

- **Events hidden by an admin** — the Hide toggle on Admin › Events removes an event from this page (and a meet whose every event is hidden disappears entirely). Used to keep test events out of public view. Direct scoreboard links still work.
- Locked events are **visible** here — locking hides events from the Officials dashboard only; public spectators should still see the final state.
- Events with a status of `setup` and no athletes — shown with a gray indicator; spectators can see what's scheduled.

When to share this URL

- **Pre-meet** — share `/livescores` with your spectator community as the meet's public landing page.
- **Mid-meet** — share with anyone wanting to follow along.
- **Post-meet** — same URL still works; events display as Complete with final standings.

Reading a Scoreboard

The Scoreboard at `/scoreboard/<short>` is the spectator-facing live results page for a single event. It updates in real time as runs are finalized.

Mogul scoreboard

The layout is an athlete-card list. Each card shows:

Collapsed

- **Rank chip** (gold for 1st, silver 2nd, bronze 3rd, neutral for others)
- **Bib chip**
- **Athlete name**
- **Best total** (big, JetBrains Mono)
- **4-up component grid** — Turns / Air / Time / Speed

Tap a card to expand:

Expanded

- **Per-phase sections** for multi-phase events (Run 1, Run 2, Q1, F1, F2)
- **Per-judge breakdowns** for each phase:
- TL judges: each judge's component scores
- A1, A2: each Air judge's jump scores and codes
- Time judge: run time
- Winning run starred

Tap the same card to collapse. Tap a different card to switch.

Dual mogul scoreboard

Dual mogul has a **3-tab strip** at the top:

1. **MATCH** (default) — current or most-recently-completed match, with full per-judge 5-point split scores.
2. **BRACKET** — tree view of every match with winner highlighted at each step.
3. **PLACE** — ranked placement table with FFSP points (after event completion).

If no match is active, the **Match** tab falls back to the most-recently-completed match (sorted by bracket_round then bracket_position) with a "MOST RECENT COMPLETED MATCH" label. If zero matches are complete, shows a "WAITING FOR FIRST MATCH" placeholder.

The **Place** tab shows DNS / DSQ / SCR markers and DNF tags. FFSP points appear only after event finalization. See FFSP points (dual mogul).

NOW COMPETING banner

When a run is in progress, a gradient banner shows above the standings:

> NOW COMPETING — Bib #7 — Smith, John

It updates live via WebSocket `run_started`. The athlete moves into the standings once their score is finalized.

Upcoming Athletes box

Below the rank table, an **Upcoming Athletes** card lists the remaining queue for the current run / phase, ordered by run order. Columns: Order, Bib, Name. Header text reflects the active phase ("Run 1 — Up Next", "Final 1 — Up Next", etc.). For single-run events, the header is "Run 1 — Up Next".

The card is muted visually (gray-on-dark) so it reads as secondary information vs. the rank table above.

Live updates

The scoreboard listens for WebSocket events:

- `score_update` › re-fetch results
- `run_started` › update NOW COMPETING banner
- `dual_match_started` / `dual_bracket_review` / `event_finalized` › re-fetch bracket / standings / status

Plus 5-second polling as a fallback. The polling skips its update if a WebSocket message arrived in the last 4 seconds (prevents stale-poll-overwrites-fresh-WS races).

Sun Mode

Toggle in the top-right (sun/moon icon). High-contrast palette for outdoor viewing.

URL & sharing

`/scoreboard/<short>` — the short code is permanent for the life of the event. Share the link via Slack, AirDrop, email — it works on any device with a browser.

PDF downloads

Buttons at the top right of the scoreboard offer the same PDF downloads as the Officials Results tab: Event Results Summary, Check Sheet by Bib, etc. See [PDF reports].

Event not found

If you visit `/scoreboard/<bad-short>`, the page shows a "Event not found" card with the bad short code echoed back and a "View live scores" link. Doesn't silently fail.

Mobile readability

Athlete cards stack to full width on iPhone. Component grids re-flow. Expanded per-judge breakdowns scroll within the card. Tested in portrait and landscape.

Broadcast Overlay (OBS/YoloBox)

The Overlay at `/overlay/<short>` is a **1920×1080 transparent-background** broadcast surface designed for OBS, YoloBox, or any other live-streaming setup. It's the same data as the public Scoreboard but in a video-friendly layout.

How to use it

1. In OBS, add a new **Browser Source**.
2. Set the URL to `https://<your-stickit-server>/overlay/<event-short>`.
3. Set width 1920, height 1080.
4. [yes] Check **"Custom CSS"** is unset (no extra styling needed).
5. [yes] Check **"Use custom frame rate"** if your stream is non-60fps.

The body / html background is forced to `transparent !important` via `useLayoutEffect`, so OBS can composite the overlay on top of your live video feed.

For YoloBox, use its "URL" source type with the same overlay URL.

Reveal animations

Each scored run triggers a reveal animation:

1. **Run start** — the OverlayRibbon shows the new athlete's bib and name.
2. **Score finalize** — the OverlayScoreReveal animates the total in with a `cubic-bezier(0.34, 1.56, 0.64, 1)` overshoot (480ms duration, 380ms delay after stat-block fade-ins).
3. **Components** — Turns/Air/Time/Speed components fade in staggered (50ms / 130ms / 210ms / 290ms).
4. **Hide** — after a few seconds, the reveal fades back to a minimal ribbon, ready for the next athlete.

Dual mogul overlay

For dual mogul events, the OverlayDualVS shows:

- **Blue side** (left) — athlete bib + name + club.
- **VS button** (center) — circular.
- **Red side** (right) — same.

Pre-result: chips show `#<bib>` on each side.

Post-result:

- Score chip on the **outer edge** of each side panel (away from VS center).
- Athletes' bibs shown in small text under their names.
- Loser's status (DNF / DNS / DSQ) replaces score chip on that side.
- Winner side gets a **gold inset border + glow shadow**.
- Winner label appears: "1ST" / "2ND" (finals), "3RD" / "4TH" (small final), "WINNER" (non-finals).

Transparent canvas

Critical for broadcast: the overlay does NOT mount `PublicLayout` (which would set a body background). Instead it injects only CSS variables, fonts, and keyframes onto a scoped `.stickit-overlay-root` class. Body stays transparent so the underlying video feed shows through.

Hardware-encoder fallback poll

When OBS / YoloBox runs in a long-lived headless browser environment without a persistent WebSocket connection, the overlay uses a **3-second polling fallback** for score updates. The dedup logic (v1.19.01) prevents the poll from overwriting fresh WebSocket data — each poll captures a `pollStartedAt` timestamp and skips its update if a WebSocket message landed after that timestamp.

Hydration on mount

When the overlay loads (e.g., OBS just started the browser source), it hydrates from the server's current state: any in-progress run shows, the most-recently-completed run's reveal animation may play (skipping manually-entered runs).

Hiding the overlay

The Scoring tab has a compact **Broadcast Overlay: Hide / Show** control (on both the single-mogul and dual panels). **Hide** blanks the overlay's content — useful between events or during a delay — and the overlay stays blank until you click **Show** or the next run starts (a new run, score, or dual match automatically un-hides it). While hidden, the overlay's fallback polling is suppressed so a score can't flicker back on screen.

Tablet / mobile

The overlay is designed for 1920×1080. Viewing on smaller screens scales it via viewport scaling, but the layout is broadcast-first, not mobile-first.

Sun Mode

Sun Mode is **not** applied to the overlay (broadcast graphics should be consistent regardless of any operator UI choice).

Multiple events on the same broadcast

For a meet with multiple events running concurrently, use separate OBS scenes — each with its own Browser Source pointing at the relevant event's overlay URL. Switch scenes to switch events on stream.

Reference

Jump codes & DDs

Each freestyle jump has a **code** and a **Degree of Difficulty (DD)** multiplier. The DD is used to multiply the air judges' raw score. Different disciplines (mogul, dual mogul, aerials) use different DD charts; men and women have separate charts too.

Where to look up DDs

Two places in the app:

1. **Officials sidebar › Jump DDs** — opens a modal listing DDs for the selected discipline + gender (filter dropdowns at the top).
2. **Air Judge tablet** — each picked code shows its DD in the AthleteBar.

Mogul DD chart (USSS / FIS)

The chart is based on:

- **Base value** by spin family (Single 0.40/0.50, Double 0.53/0.63, Triple 0.66/0.76, Quad 0.79/0.89, Quint 0.92/1.02 — men/women).
- **Modifiers per letter** added to the base:
- `T` (twist) › -0.02
- `S` (spread) › -0.02
- `D` (daffy) › +0.01
- `X` (X-out) › +0.01
- `Y` (y-spread) › +0.01
- `M`, `K` › +0.01 each
- `Z` › 0
- `g` (basic grab) › +0.05
- `G` (advanced grab) › +0.12

Basic vs advanced grabs (v1.26.00, FIS JH 6204.3.7)

Since the Spring 2026 FIS rule change, moguls distinguishes a **basic grab** (lowercase `g`, modifier +0.05) from an **advanced grab** (uppercase `G`, modifier +0.12 — previously +0.14). `bg` and `bG` are **different jumps with different DDs**, so jump-code entry is case-exact. Examples (men / women):

Code	DD (M / F)	What it is
`3g`	0.73 / 0.83	360 with basic grab
`3G`	0.80 / 0.90	360 with advanced grab
`bg`	0.73 / 0.83	Back flip with basic grab
`bG`	0.80 / 0.90	Back flip with advanced grab
`g`	0.45 / 0.55	Stand-alone basic grab

Code	DD (M / F)	What it is
`G`	0.52 / 0.62	Stand-alone advanced grab

Because only identical codes are repeats under USSS 4210.2.1, `bg` and `bG` are **not** repeats of each other. The FIS Moguls Advisory Group will conduct a full DD chart review at the start of the quad; these values implement the approved FS-13 modifiers, and individual values remain editable through the Admin › Jump DDs table.

Common codes and their DDs (men / women):

Code	DD (M / F)	What it is
`S`	0.38 / 0.48	Single spread
`T`	0.38 / 0.48	Single twist
`TS`	0.49 / 0.59	Double twist with spread
`TT`	0.49 / 0.59	Double twist (helicopter, 720)
`TD`	0.52 / 0.62	Double twist with daffy
`TTS`	0.62 / 0.72	Triple twist with spread
`3`	various	Triple twist (1080)
`3p`	various	Triple twist position (varies)
`bp`	varies	Back layout position (back family)
`bT`	varies	Back twist
`bL`	varies	Back layout
`bF`	varies	Back full (back tuck flip)

The full chart is seeded into the `jump_dd_table` database on server boot. v1.16.08 corrected several values; if your server pre-dates that, a startup migration auto-corrects.

Aerials DD chart (USSS Appendix C)

A separate chart for aerials, sourced from the 2026 USSS Freestyle Competition Guide page 99. 29 base entries plus 42 spin-family combinations programmatically expanded.

Key examples:

Code	DD
`S`	1.48
`D`	varies
`T`	varies
`bL`	2.05
`bF`	2.30
`bFF`	3.15
`bdFF`	3.525

Dual mogul DDs

Dual mogul uses the **mogul chart × 1.25**. The modal labels this clearly: "Dual Mogul (x1.25 applied)".

Per-jump scoring

For mogul (v1.24.00, per FIS JH 6203.2.2 / 6204.3):

- Each air judge enters a per-jump raw score (0–10).
- The judges' raw scores are averaged per jump, truncated to 2 decimals, then multiplied by the DD.
- Each jump is capped at 10.0 points.
- Summed across jumps (or doubled for 1-jump events), capped at the 20-point air max per FIS JH 6204.

For aerials (v2):

- Each scoring judge enters Air, Form, Landing per jump.
- Reduction (drop high+low or operator-selected) applied per component per jump.
- `floor((sumKept(Air) + sumKept(Form) + sumKept(Landing)) × DD, 2dp)` per jump.
- Summed across jumps.

Where DDs come from

The `jump_dd_table` rows are seeded on server boot (`seedDualMogulDDs`, `seedAerialsDDs`, `seedJumpDDs` for moguls). A sentinel check on startup detects stale charts and re-seeds if needed (e.g., the v1.16.08 mogul fix or the v1.18.02 aerials chart correction).

Calling the API directly

`GET /api/jump-dds?ruleset=uss&discipline=mogul&gender=M` returns the full chart as JSON. Used by the Officials sidebar's Jump DDs modal and by the voice modal's jump code validation.

Scoring formulas

Every score in StickIt comes from a published FIS / USSS formula. This reference lists them in one place. All values are truncated (floored) to 2 decimals per FIS rule; DDs are preserved at full precision.

Mogul total

```
total = turns + air + speed
```

Capped at 100.0. Truncated to 2dp.

Turns (max 60)

5-judge format (3 counting + 2 reference): sum the 3 counting T&L judge scores. **7-judge format (5 counting + 2 reference):** drop the high and low **turns** scores and the high and low **deductions** independently, then sum the kept values (per FIS JH 6203.1.1 — the drops are per column, not per judge's net).

Per FIS JH 6203.

Per-judge total:

Component scoring on (Comp Series, FIS):

```
judge_total = clamp(carving + absorption + upper_body - deduction, 0.1, 20)
```

Component scoring off (RQS-EQS):

```
judge_total = clamp(raw - deduction, 0.1, 20)
```

Air (max 20)

Per FIS JH 6203.2.2 / 6204.3 (order matters — average first, then DD):

```
per_jump_air = floorToHundredth(avg(judge raw scores)) × DD, capped at 10.0
air_total = floorToHundredth(jump1_air + jump2_air)
```

A jump whose averaged score is below 0.1 earns no points. If only 1 jump (Devo), double the single-jump value:

```
air_total = floorToHundredth(2 × per_jump_air)
```

Capped at 20.0. Single-jump-in-2-jump-event capped at 10 per USSS 4210.2.2.

Repeated jumps

When an athlete performs the same jump twice in a run, only one counts:

- **RQS-EQS:** the higher-**scoring** jump counts; the lower-scoring jump is zeroed (tie › jump 1 counts).

- **All other divisions:** the first jump counts per USSS 4210.2.1; the second is zeroed.

The dropped jump is also excluded from the Air-no-DD tie-break value.

Air-no-DD (tie-break only)

```
air_no_dd = avg(per-judge air raw, jump 1) + avg(per-judge air raw, jump 2)
```

Floored to 2dp. Single-jump events double the one jump's mean. No DD, no cap. Used in mogul and aerials tie-break per FIS ICR 4207.3 and USSS 4110.4.3.

Speed (max 20)

```
speed = max(0, 48 - 32 × (run_time / pace_time))
```

Floored to 2dp. Capped at 20.0. Per USSS / FIS ICR 4206.3.

If `run_time <= 0` (e.g., NT sentinel `-1`), speed = 0.

Pace time

```
pace_time = floor(course_length / pace_speed, 2dp)
```

Pace speeds by standard:

- USSS: Men 9.70 m/s, Women 8.20 m/s
- FIS: Men 10.30 m/s, Women 9.00 m/s

Per USSS 9.70 / 8.20 m/s; FIS 10.30 / 9.00 m/s.

Aerials v2 (per jump)

```
total_judges_score = sumKept(Air) + sumKept(Form) + sumKept(Landing)
jump_score = floor(total_judges_score × DD, 2dp)
event_total = sum across jumps (floored to 2dp)
```

Reduction:

- 5+ judges › drop high + low per component independently
- 2–4 judges › operator's choice: `sum_all` / `drop_high` / `drop_low` / `average`

Per FIS JH 6004 / USSS 4110.

Aerials legacy (pre-v1.18.00)

Single Form / Landing per run, Air-only DD multiplication. Not used for new events. Historical results render unchanged.

Dual mogul (5-point split)

Each judge awards 5 points split between blue and red (e.g., `4-1`). Match winner = athlete with majority across the 5 judges.

Tie-break per FIS:

- Components: Turns, Air, Time, Overall (judge role determines weight)
- Then re-ski if still tied

Tournament placement via `server/dual/placement.js` (band-based randomization for seeded brackets).

Manual time calculation

```
total_time = top_time + bottom_time
```

Used when no full-course timer is available. The bottom/top times are summed and treated as the run time in the speed formula.

FFSP (dual mogul only)

```
tier = is_divisional › [1000, 970, 950]
     : CC >= 15      › [900, 875, 855]
     : CC < 15      › [800, 775, 760]

points(1) = tier[0]
points(2) = tier[1]
points(3) = tier[2]
points(n) = max(0, tier[2] - (n - 3) × (tier[2] / CC))   for n >= 4
```

DNS / DSQ / scratched › 0 points, excluded from CC. First-round DNF › 0 points.

Mogul tie-break (FIS ICR 4207.3)

1. Total
2. Turns
3. Air-no-DD
4. Speed

Each compared with 0.001 epsilon. Ties at all four levels › shared rank with Olympic-style skip.

Aerials tie-break (USSS 4110.4.3)

1. Total
2. Air-no-DD
3. Form
4. Landing

Same epsilon and skip-rank rule.

Truncation rule

All published numeric values are **floored** to 2 decimals. This applies to:

- `runs.total_score`
- `runs.turns_score`
- `runs.air_score`, `runs.air_score_no_dd`
- `runs.speed_score`
- per-jump air values
- speed score from a course-length recompute
- pace time

DDs are preserved at full precision.

Viewer API (third-party apps)

StickIt exposes a read-only public API at `/api/viewer` for iOS apps and other third-party clients that want to display live competition results. No authentication is required — it stays open even when password protection is on — and it never exposes anything a spectator couldn't already see on the public Scoreboard.

Endpoints

Endpoint	Purpose
`GET /api/viewer/events`	Event list across all meets; `?status=in_progress` filters to live-only
`GET /api/viewer/resolve/:shortCode`	Look up an event from its short code (the same code used in `/scoreboard/<short>` URLs)
`GET /api/viewer/events/:id/status`	Live state: current round, athlete on course, upcoming athletes (10 by default; `?upcoming_limit=` up to 100, v2.2.00)
`GET /api/viewer/events/:id/results`	Ranked results for the active round — or any round via `?run_number=` (v2.2.00) — for mogul / aerials; full bracket state for dual mogul
`GET /api/viewer/events/:id/results/scores`	Per-judge raw scores for the active round or a requested `?run_number=` — for expanded detail views
`GET /api/viewer/events/:id/results/phases`	Combined standings for phased events (Best of 2, Qualifier/Finals) with tiers, per-run detail, and the counting run marked (v2.2.00)
`GET /api/viewer/events/:id/rounds`	Round / phase list with status (not started / in progress / complete)
`GET /api/viewer/events/:id/dual-matches/:matchId/judge-points`	Per-judge blue/red point splits for one dual match (v1.26.00)
`GET /api/viewer/events/:id/placements`	Full-field ranked final standings for a dual mogul event, per ICR 4312 — matches the web PLACE tab (v1.30.00)

Full request/response examples live in the project README's **Viewer API Reference** section.

Behavior notes

- **Hidden events are excluded from the event list.** Events hidden via Admin › Events don't appear in `GET /api/viewer/events`, but the per-event endpoints (resolve, status, results, rounds) still work for anyone who has the short code — same listings-only rule as the Live Scores page.
- **Results follow the active round, or a requested one.** Between rounds, the most-recently-completed round's results are returned, so a client always has something to show. As of v2.2.00, `?run_number=` on `/results` and `/results/scores` requests a specific round (unknown rounds get `400 { "error": "Unknown run_number" }`), and the response echoes the `run_number` actually served. As of v1.30.00 each mogul/aerials result row carries its `registration_id`, so per-judge detail from `/results/scores` can be joined by id instead of by list position.
- **Ranking is rule-correct (v2.2.00).** Mogul/aerials results are ordered by the engine's FIS tie-breaks; tied athletes share a rank with Olympic-style skips, and DNS/DNF/DSQ/RNS athletes are placed per USSS 4012.3 (scored › DNF › RNS › DNS, DSQ at the bottom) with `effective_status` set. Clients should render rows in array order rather than re-sorting by rank.

- **Phased standings** (`.../results/phases`, v2.2.00) return the same combined view as the web Scoreboard for Best of 2 and Qualifier/Finals events — tier labels, shared ranks, a per-run map keyed by run number, and `counts: true` on the run that ranks each athlete. Dual mogul events get `400`; unphased events return `format: "none"`.
- **Dual mogul** results return the full bracket with blue/red athletes, per-side scores, winner, and bye flags — enough to render a bracket tree. As of v1.26.00 each match also carries its `id` (use it with the judge-points endpoint). As of v1.29.00 each match carries `nj_call` (`null` / `"blue"` / `"red"` / `"both"` — FS-18 landing zone No Jump call) and `blue_score` / `red_score` are **effective** totals (NJ override + tie credits applied — a tied-time match totals 25, matching the web scoreboard). As of v1.30.00 each match also carries `winner_side` (`"blue"` / `"red"` / `null`) — the authoritative winner from the stored winner id, correct even for NJ-decided and tie-break matches where the point totals alone would mislead — plus `registration_id_blue` / `registration_id_red`.
- **Dual placements** (`.../placements`, v1.30.00) return the entire field ranked per ICR 4312 — `{ discipline: "dual_mogul", placements: [{ rank, registration_id, bib_number, gp, first_name, last_name, club, run_status, reg_status }] }`. `rank` is `null` for unclassified entries (first-round DNS in seeded groups, DSQ); `bib_number` is a number or `null`. Non-dual events get `400`.
- **Judge points** (`.../dual-matches/:matchId/judge-points`) returns, as of v1.29.00, `{ match_id, nj_call, speed_tied, air_tied, overall_scale, judges: [...], effective_judges: [...] }`. `judges` are the raw entries (`judge_number, blue_points, red_points, time_tied, air_tied`); `effective_judges` are the values that actually pay (NJ-overridden Time Judge row, 3/3 speed-tie credit, 0/0 air-tie, `overridden` flag) — display these so the app never has to recompute rules. A match id that doesn't belong to the event returns `404 { "error": "Match not found" }`.
- Unknown event IDs or short codes return `404 { "error": "Event not found" }`.

Sharing with a developer

Give an app developer two things: your server's base URL and an event short code (visible in any scoreboard link). Everything else is discoverable from the endpoints above.

Glossary

A quick reference for the acronyms and terms that show up throughout StickIt.

Organizations & rule books

USSS — US Ski & Snowboard. The national governing body for freestyle skiing in the US. Sanctions Comp Series, RQS-EQS, Divisional Championships, and Devo events.

FIS — Fédération Internationale de Ski. The international governing body. Sanctions FIS-rated events including World Cup, Continental Cup, NorAm, and FIS-other.

ICR — International Competition Rules. The FIS rule book for freestyle skiing. Section 4207 covers tie-breaks; 4205 covers run order; 4206 covers speed; 4210 covers air.

JH — Judging Handbook. The FIS-published companion to ICR with detailed judging procedures. JH 6003 covers aerials reduction; JH 6004 covers per-judge-per-jump scoring; JH 6203 covers mogul turns drop-high/drop-low.

RMF — Rocky Mountain Freestyle. The Rocky Mountain Division of USSS.

Disciplines

Mogul — Single athlete down a moguled course, scoring Turns + Air + Speed. The default discipline.

Dual Mogul — Head-to-head bracket format. Two athletes ski parallel courses simultaneously; 5-point split scoring.

Aerials — Single athlete launching off a kicker. Score Air + Form + Landing per jump.

Judge roles

T&L — Turns & Line. The judges scoring the moguls run for skiing technique. Common roles: TL1, TL2, TL3.

Air — Air judges. Score the jumps in a mogul run.

HJ — Head Judge. Reviews and approves submissions, finalizes.

Timekeeper — Records run time. Mogul / aerials only (dual mogul has its own DualTime judge).

AeJudge — Aerials Judge (v2 model). Numbered AeJudge1 through AeJudge<N>.

Score components

DD — Degree of Difficulty. A multiplier applied to air judges' raw scores per jump.

Turns — Turns & Line score component (max 60 for mogul).

Air — Air score component (max 20 post-DD for mogul; per-jump for aerials).

Air-no-DD — Pre-DD raw air execution. Used in tie-break only.

Speed — Speed score component (max 20 for mogul).

Form — Form score (aerials, 0–5 per judge per jump).

Landing — Landing score (aerials, 0–3 per judge per jump).

Time Points — Speed score, alternate name used in some UI surfaces.

Status codes

DNS — Did Not Start.

DNF — Did Not Finish.

DSQ — Disqualified.

RNS — Refused to be Scored. Legacy status retired in v1.26.00; historical values still display and transmit to USSS as DNF.

NT — No Time. Mogul / aerials only. Speed score = 0.

SCR — Scratched. Registration status, not run status.

NJ (dual mogul) — No Jump on the bottom air: the competitor landed past the Landing Zone ("chop", max 20 m from takeoff) per FS-18. Called by the Air Judge (HJ can set/clear). One NJ = 0 speed points to the violator / 5 to the opponent; both NJ = speed tied at 3 / 3. Distinct from the mogul jump code `NJ` (DD 0.00) used on the air-judge tablet. See [Dual mogul setup].

Time Tied / Air Tied (dual mogul) — tied comparisons declared by the Time Judge (J4) and Air Judge (J3). A speed tie is credited 3 / 3 (five-judge panel); an air tie withholds the votes at 0 / 0. Each tie drops the Overall Judge's split by one point (5 › 4 › 3).

Basic / advanced grab — since v1.26.00, lowercase `g` is a basic grab (DD modifier +0.05) and uppercase `G` an advanced grab (+0.12). `bg` and `bG` are different jumps. See [Jump codes & DDs].

Phases

Run 1 / Run 2 — Best of 2 mogul format.

Q1 / Q2 / F1 / F2 — Qualifier 1 / 2, Final 1 / 2. Used for qualifier/finals events.

CC — Counting Competitors. Used in FFSP formula. Total placements minus DNS / DSQ / scratched.

Categories / divisions

Comp Series — Standard USSS competition tier.

Devo — Developmental events for younger athletes. 1 jump, no time, simpler scoring.

RQS-EQS — Regional / Eastern Qualifying Series. Uses non-component T&L scoring.

Divisional — A divisional-championship-level event (`events.is_divisional=1`); top FFSP tier.

Aerials tiers

FIS OWG / WSC / WC — FIS Olympic Winter Games / World Ski Championships / World Cup.

FIS NAC/NorAm — FIS Continental Cup / North American Cup.

FIS Other — Other FIS-sanctioned events. Locked at 5-judge panel.

USA National — USSS National-level event.

USA Regional — USSS Regional-level event. HJ may score (counts as a panel judge).

Scoring concepts

FFSP — Freestyle Skiing Points. USSS point system. StickIt computes for dual mogul only.

Floor to 2dp — Truncate to 2 decimal places per FIS rule. e.g., `floorToHundredth(7.4567) = 7.45`.

Olympic-style skip rank — When N athletes share a rank, the next rank skips by N. Tied 3rd › next is 5th, not 4th.

Pace time — `course_length / pace_speed`, floored to 2dp. Used in the speed score formula.

Pace standard — `USSS` (default) or `FIS`. Determines the pace speeds.

Technical

Short code — A short random string used in public URLs (e.g., `/scoreboard/abc12`). Each event and each judge has its own short code.

Tablet URL — The unique URL a judge opens on their tablet (e.g., `/judge/<event-short>/<judge-short>`).

WebSocket — Server-pushed updates over `/ws`. Scoreboards, tablets, and overlays all subscribe.

Polling fallback — A periodic GET request used as a fallback when WebSocket is dropped.

HC mode — High-contrast mode on tablets. Black/white/amber palette for outdoor visibility.

Sun Mode — High-contrast mode for public surfaces (Home, Live Scores, Scoreboard). Like HC mode but for spectators.